
Transformers for NLP & Large Language Models

— Intro to AI - Autumn 2025 Class —

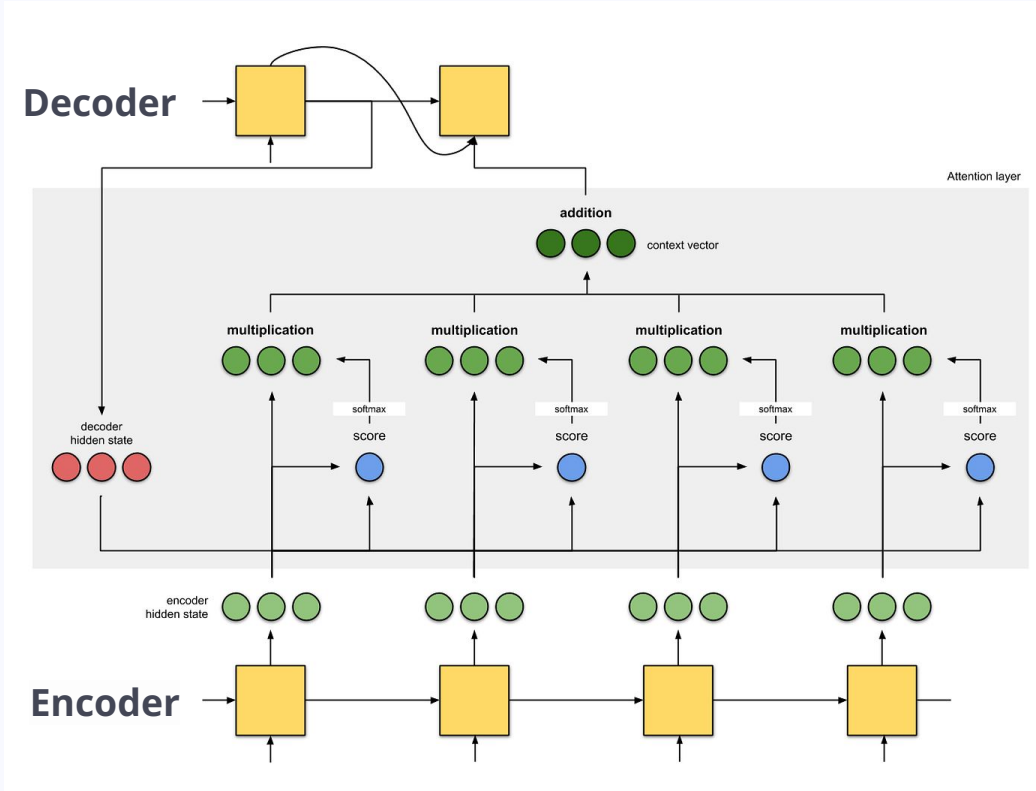
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 - c. Language Generation

1. Self-attention

Were you paying *attention* last week?



Self-attention

Traditional attention mechanisms are good... but dependent to current model!

What if attention were independent...

Self-attention: type of attention in a Transformer NN ([Vaswani et al., 2017](#))

Instead of encoder-decoder hidden states, we compare Keys:Values - Queries

Key:Value - Query

Inspiration from Information Retrieval

Key	Value
Marchand	Léon
Riner	Teddy
Lebrun	Félix
Laurin	Althéa
Brunet	Manon

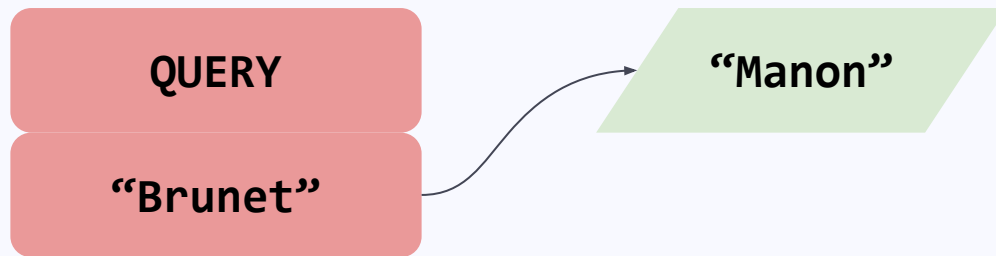


Key:Value - Query

Inspiration from Information Retrieval

Key	Value
Marchand	Léon
Riner	Teddy
Lebrun	Félix
Laurin	Althéa
Brunet	Manon

There is an exact match in the database!

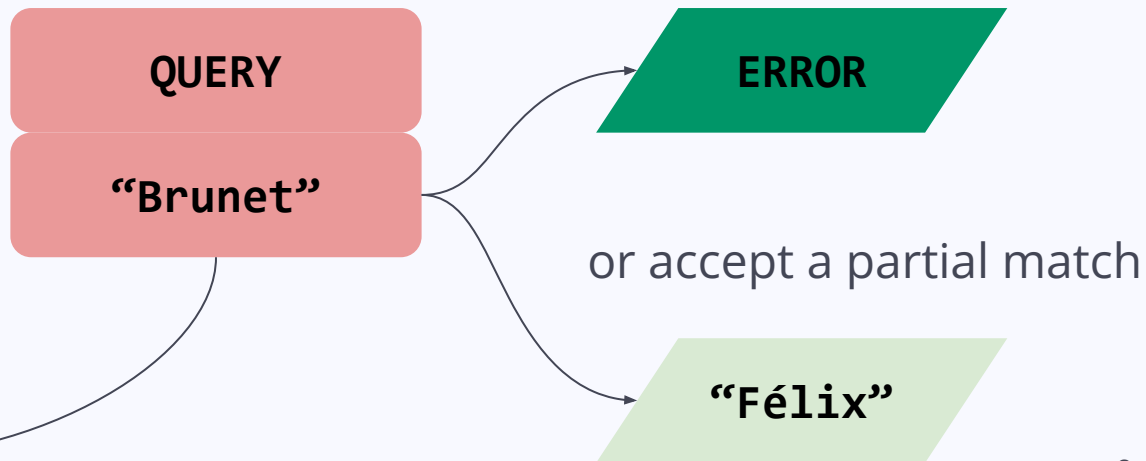


Key:Value - Query

Inspiration from Information Retrieval

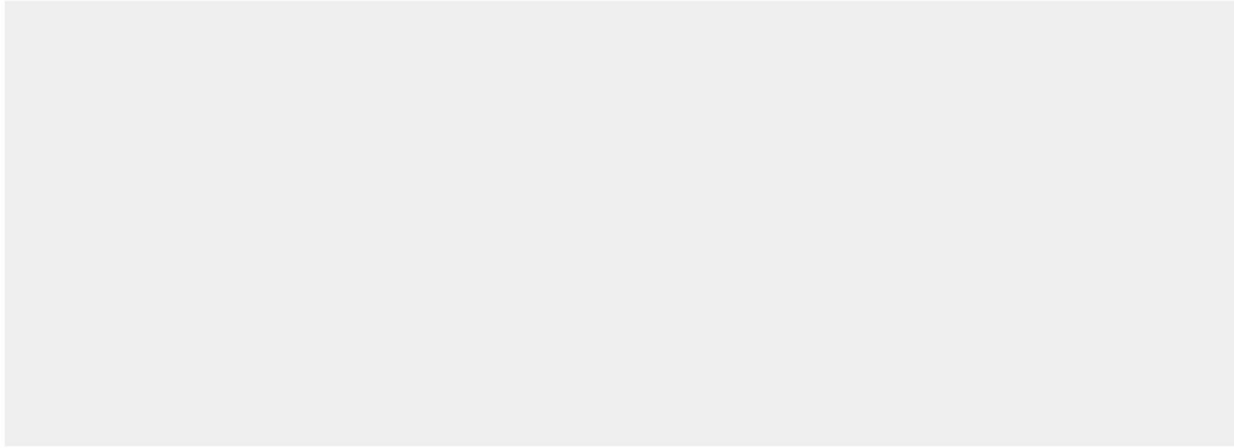
Key	Value
Marchand	Léon
Riner	Teddy
Lebrun	Félix
Laurin	Althéa
Vaast	Kauli

There is no exact match in the database!



Self-attention steps

Self-attention



input #1

1	0	1	0
---	---	---	---

input #2

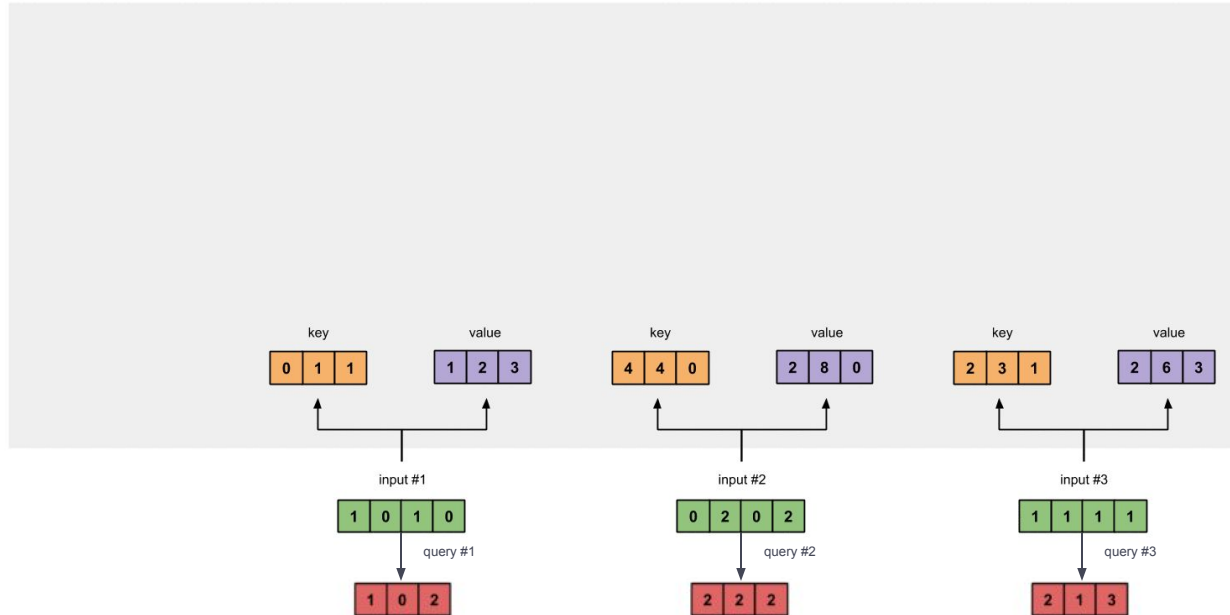
0	2	0	2
---	---	---	---

input #3

1	1	1	1
---	---	---	---

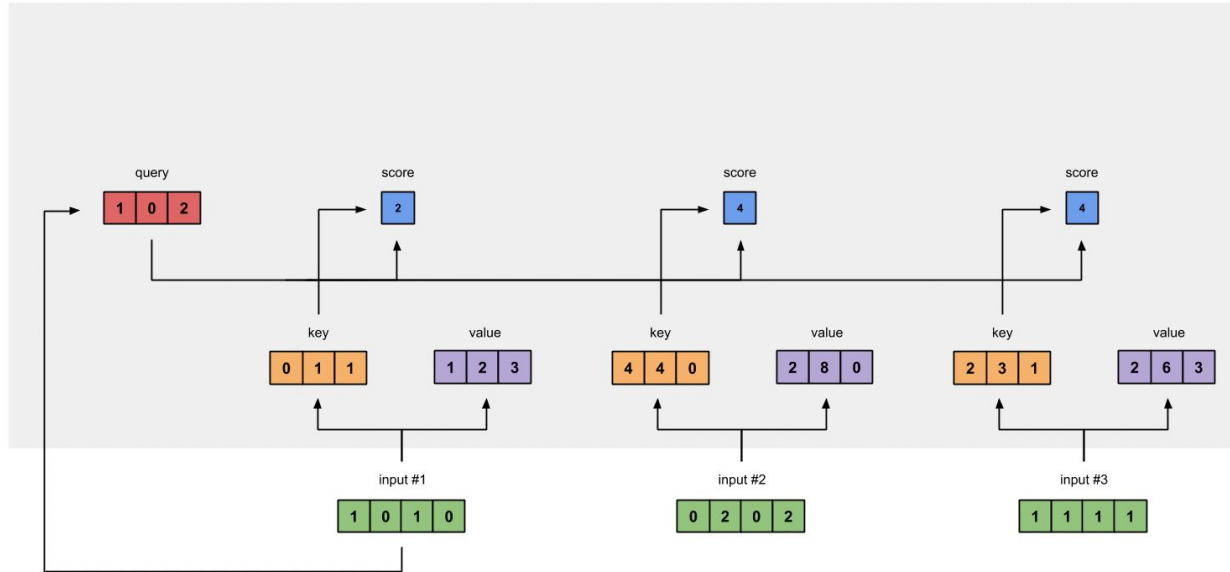
Self-attention steps

Self-attention



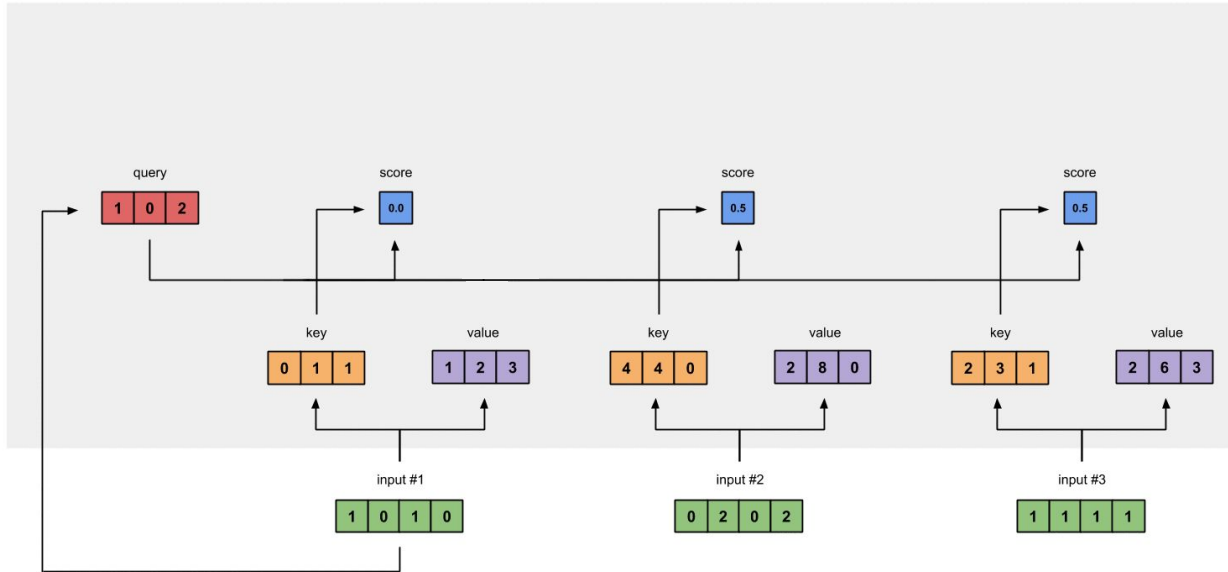
Self-attention steps

Self-attention



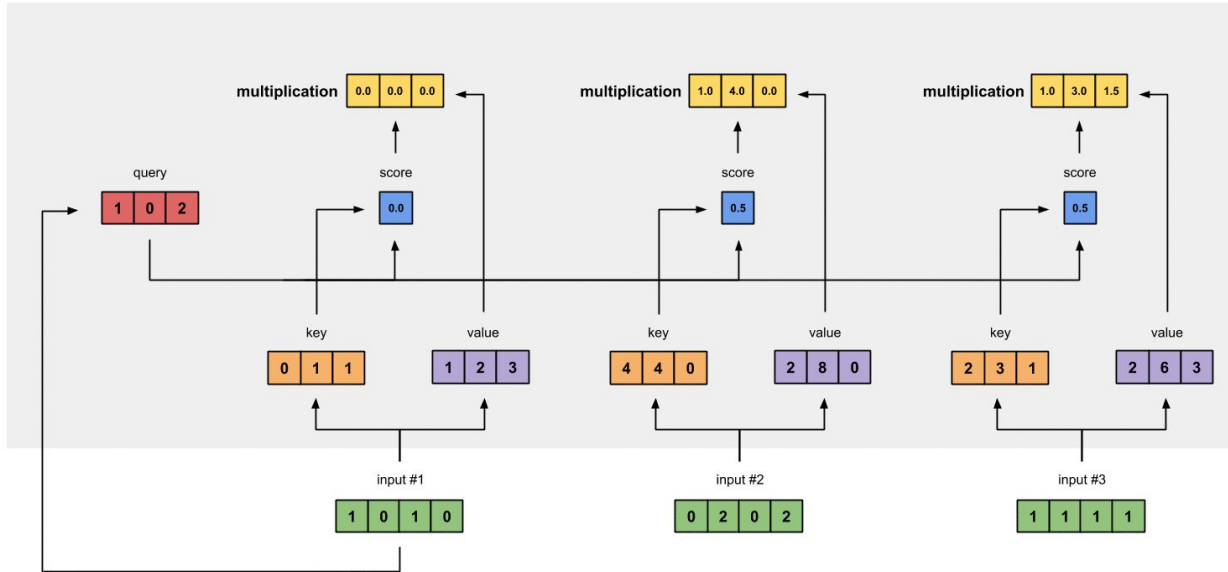
Self-attention steps

Self-attention

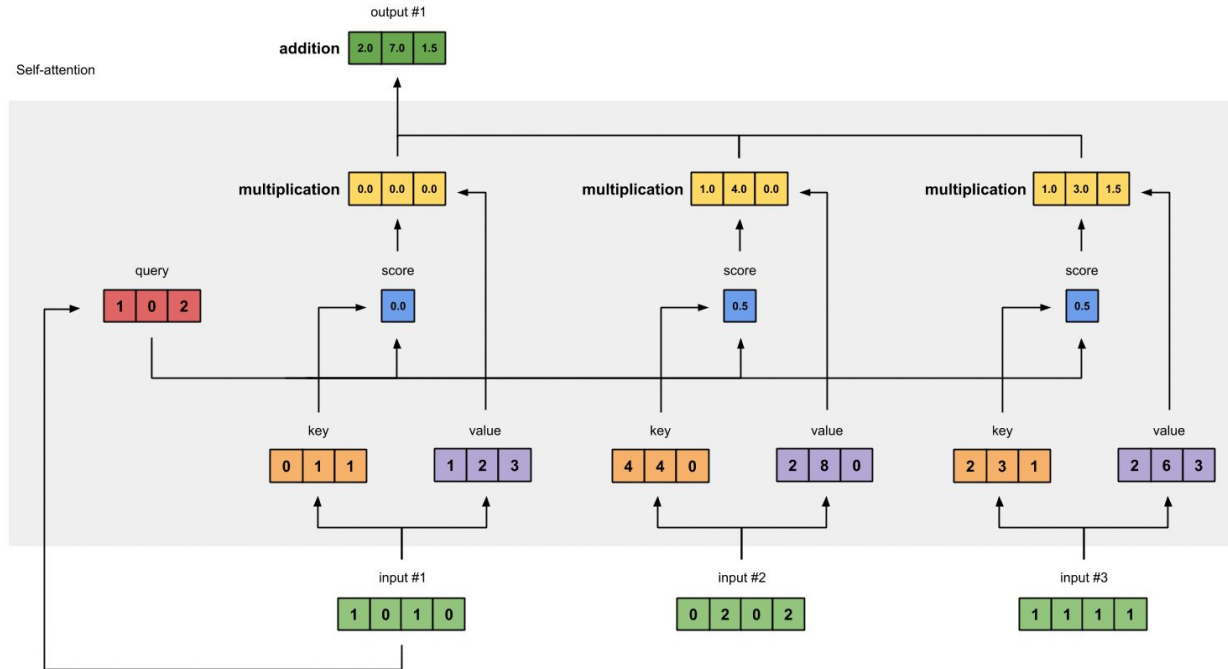


Self-attention steps

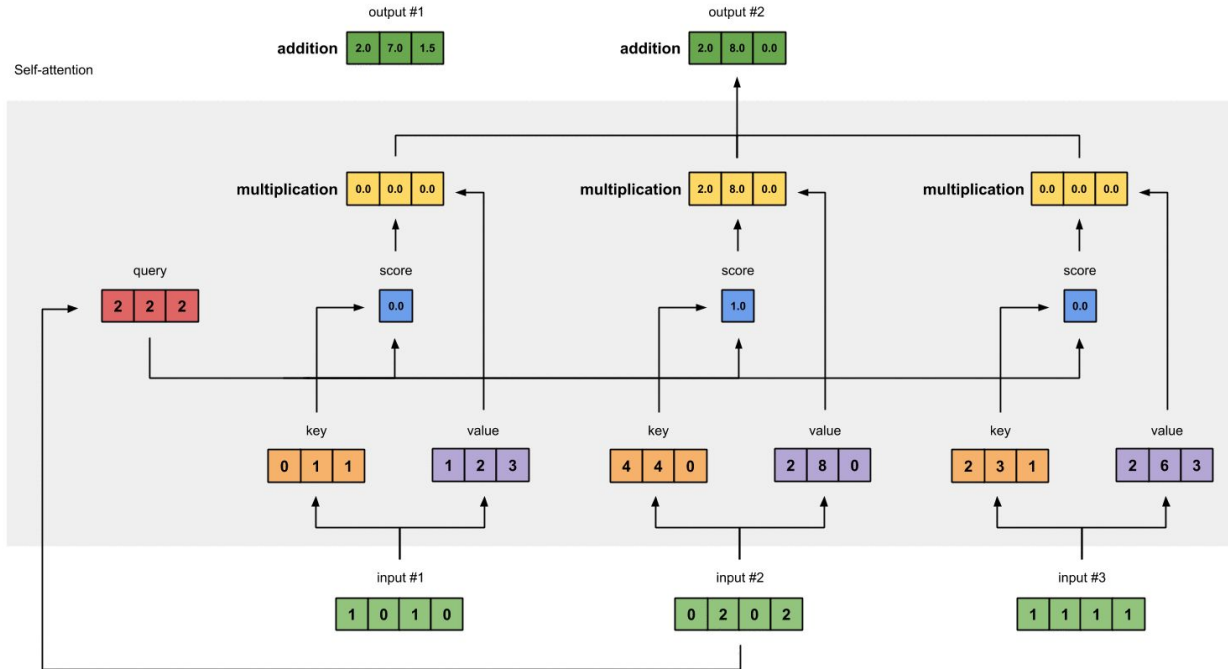
Self-attention



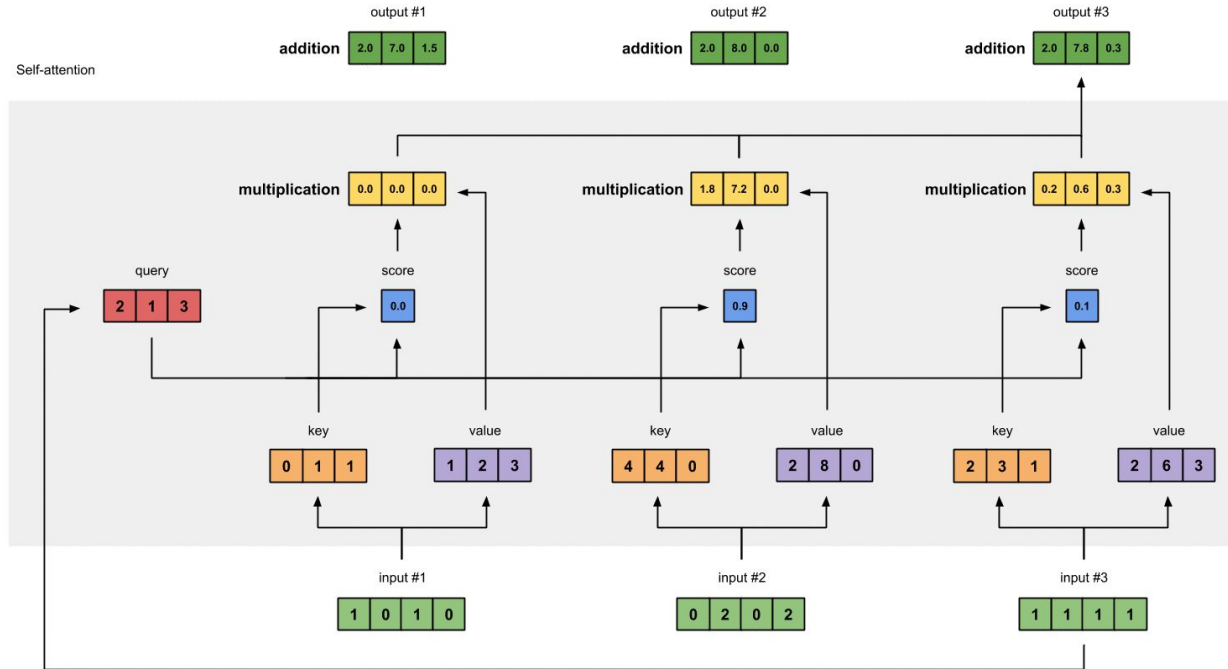
Self-attention steps



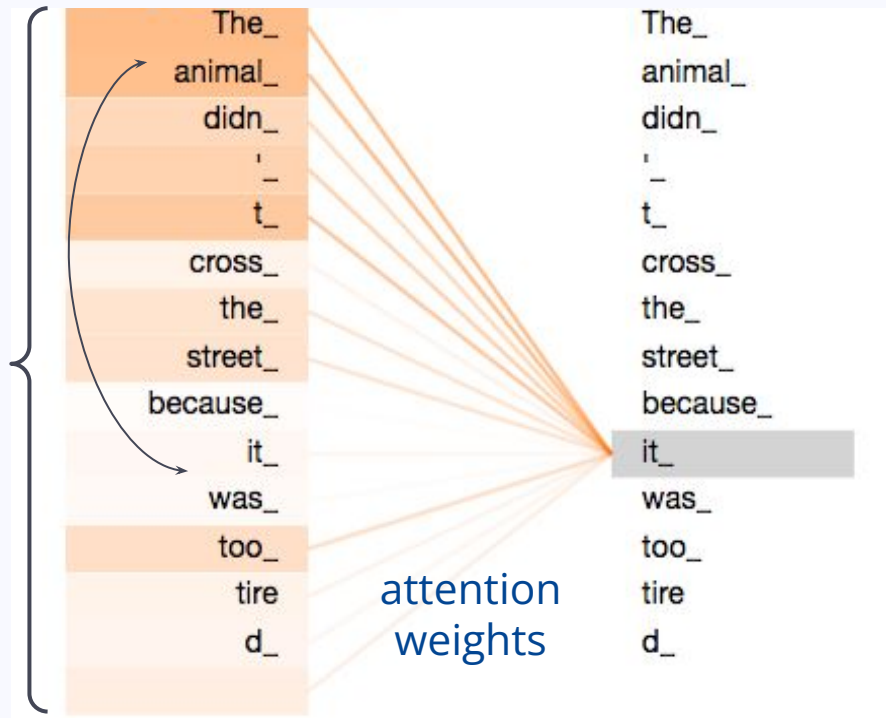
Self-attention steps



Self-attention steps



Self-attention representations



Attention of **token “it”**:

- representations for all tokens (+ itself)
- attends to “the”, “animal” the most
- also captures negation (important!)

Self-attention: Pros & Cons

- + Great at **long-distance relations!**
- + Good at **coreference resolution!**
- + Better **contextual understanding & representations**
- + **Parallel computations** for efficiency and scalability
- Very **computationally complex!**
- Needs multiple **layers** and **heads** to be really good

2. Transformers

The path to Transformers

seq2seq models



learning input
serially

The path to Transformers

seq2seq models → seq2seq + attention



learning input
serially



learning input serially &
learning important
“shortcuts” of context

The path to Transformers

seq2seq models → seq2seq + attention → Transformers + self-attention



learning input
serially



learning input serially &
learning important
“shortcuts” of context



parallelized learning &
use of self-attention for
word representations

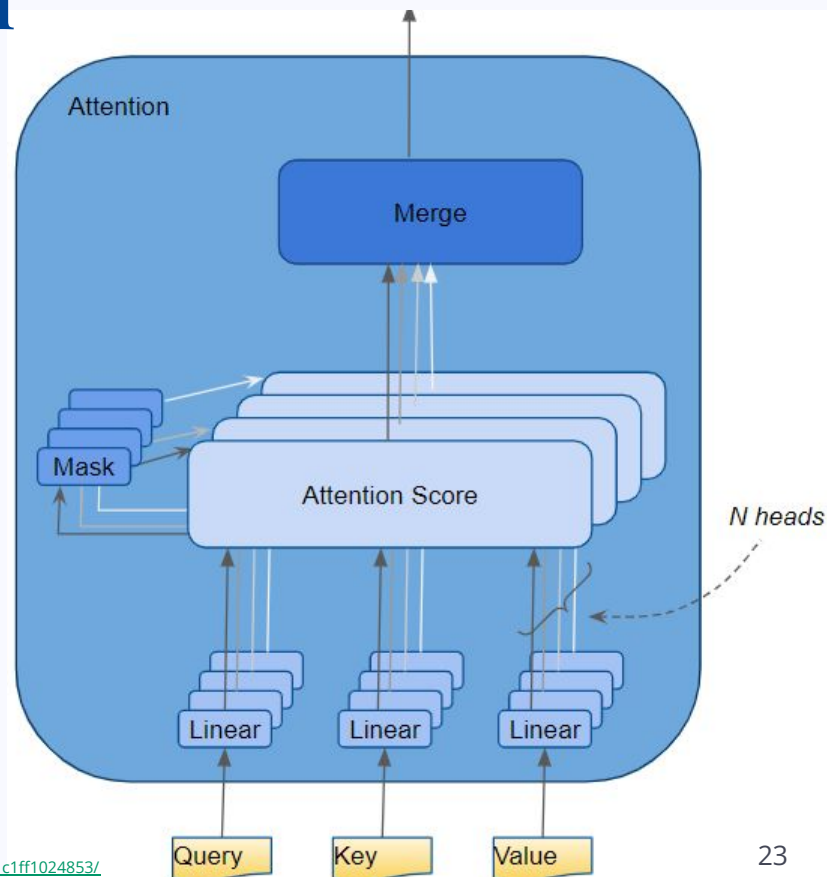
collects attention from the entire input,
creates **representations**
(+ **multi-headed**, i.e. many subspaces!)



Multi-head Self-attention

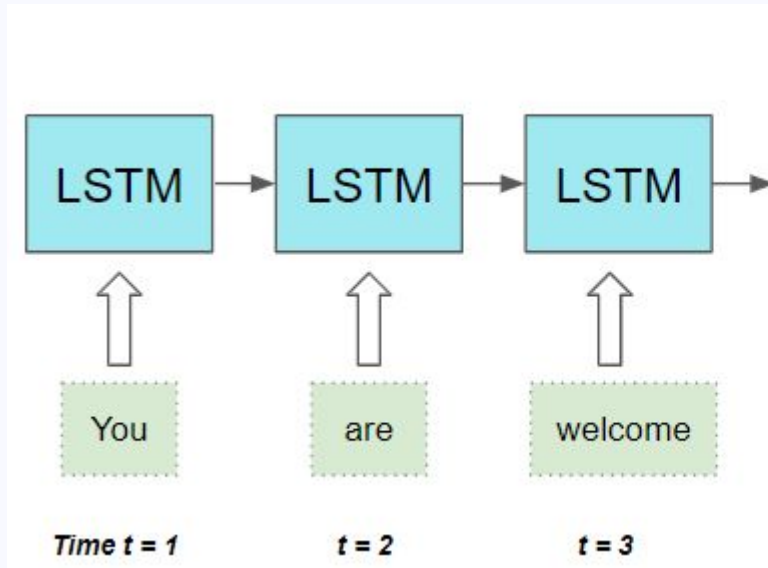
Inside the encoder, not between enc-dec!

1. Start with keys:values - queries
2. **Split** them in N parts ($N = \#$ of heads)
3. **Each head** computes
4. **Combine** heads' attention scores →
final attention score

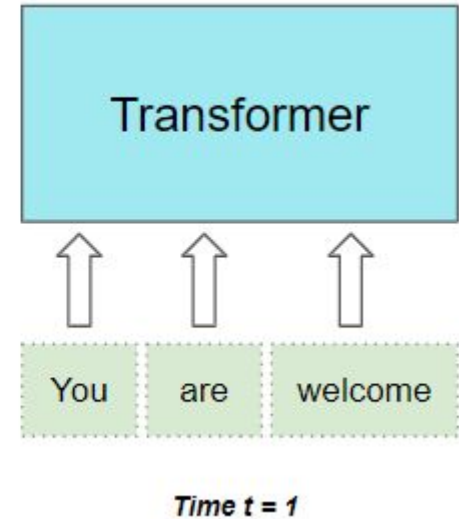


The path to Transformers

Traditional NNs: Multiple timesteps

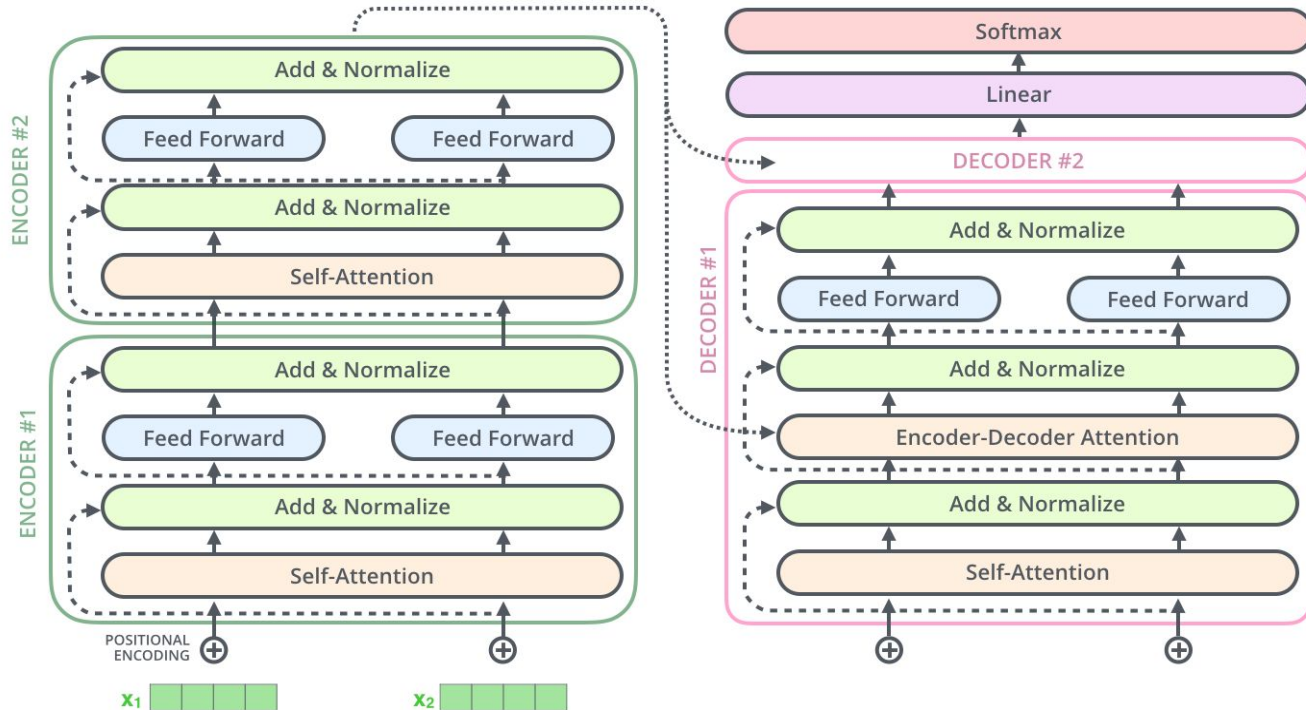


Transformers: Parallel!



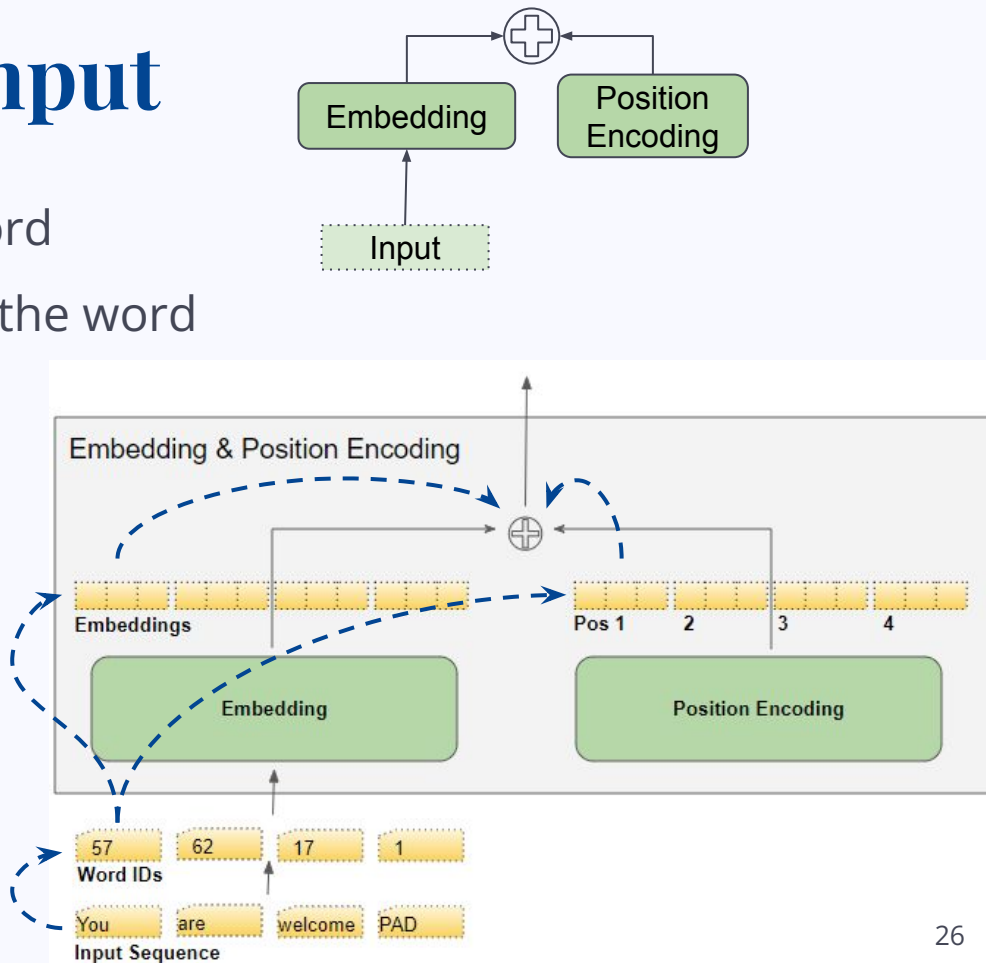
What is a Transformer?

NN with a stack of **Encoder & Decoder layers** using **multi-head self-attention**



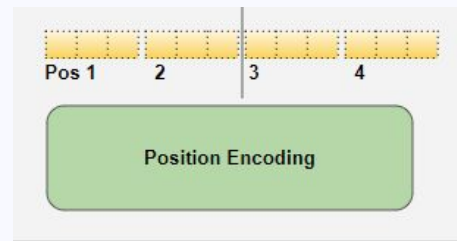
The Transformer's Input

- **Embedding** = meaning of the word
- **Position encoding** = position of the word
- Transformer **combines** them!

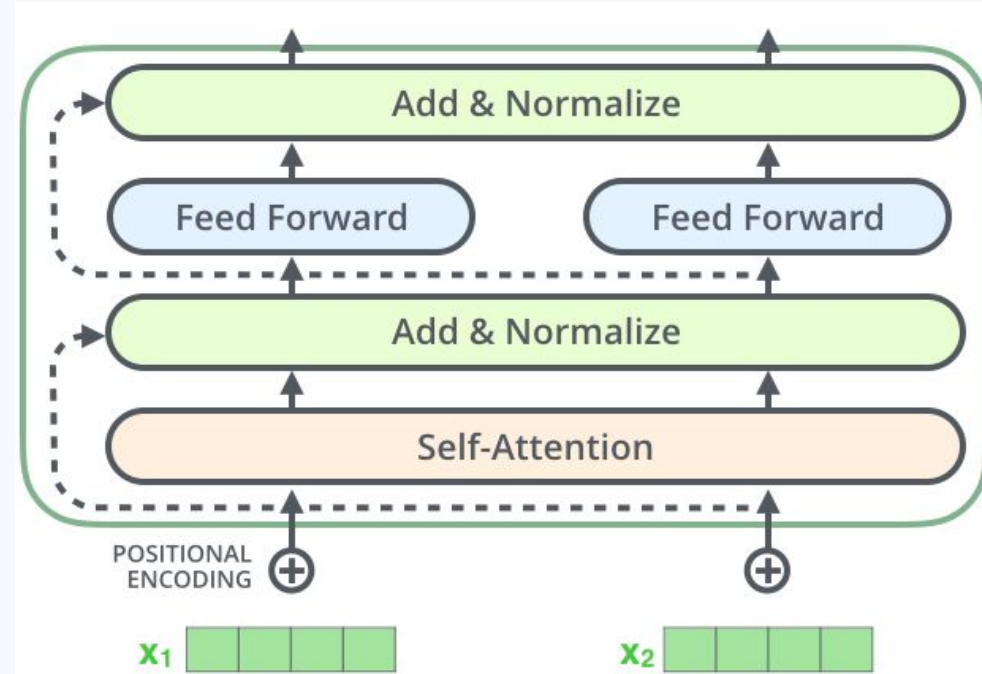


Positional Encodings

- **Traditional NNs:** sequential processing → position is evident!
- But with parallel processing in **Transformers**?
- **Positional encoding:** an independent layer from the input sequence
- Encodes the position, length and index value of token
- More complex than just $[0, 1, 2, 3, \dots]$...

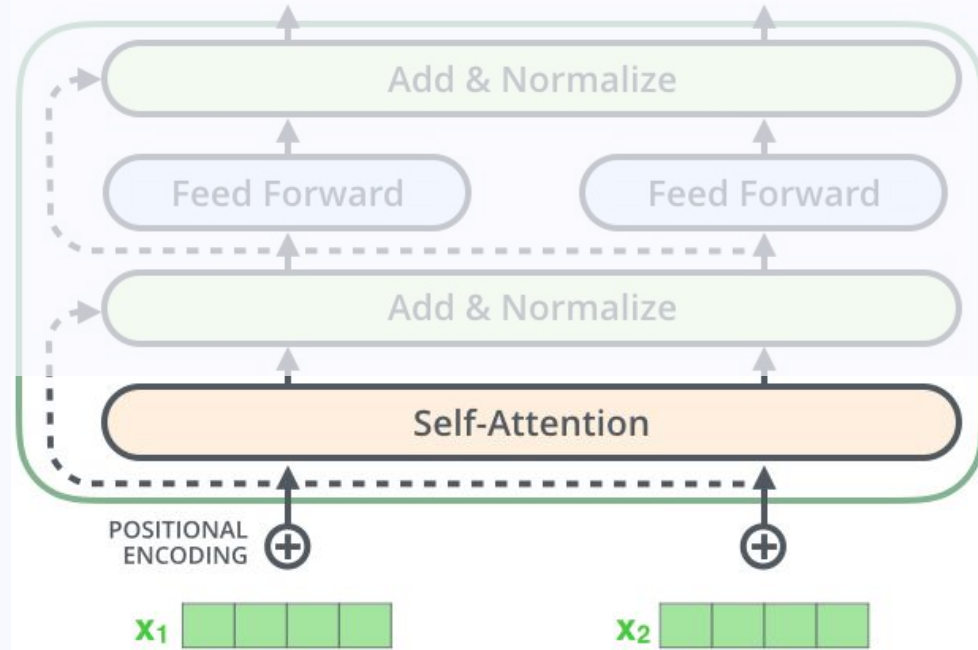


The Transformer's Encoder #1



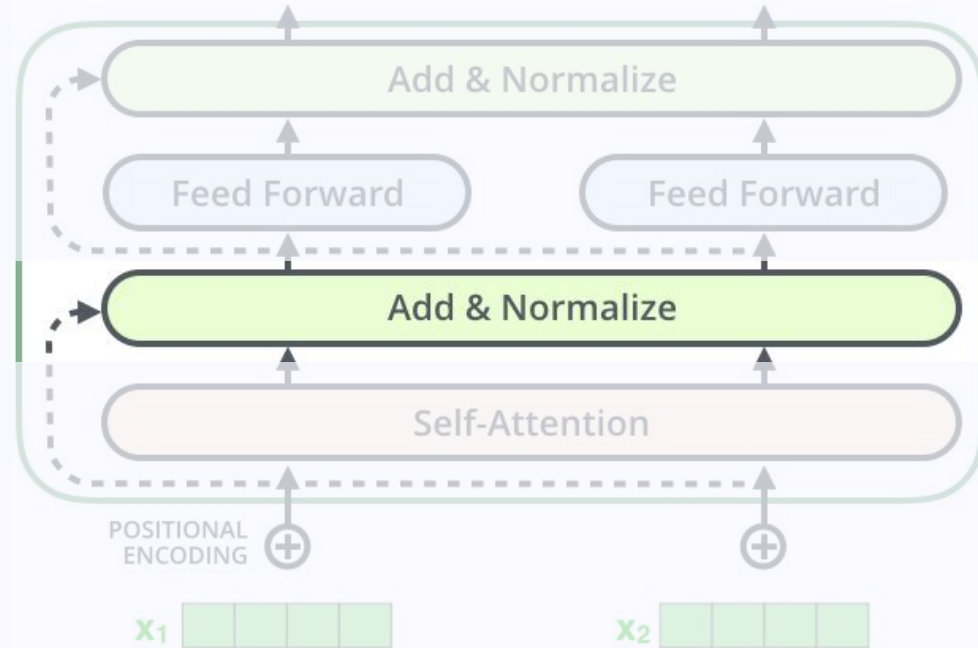
The Transformer's Encoder #1

1. Multi-headed Self-attention



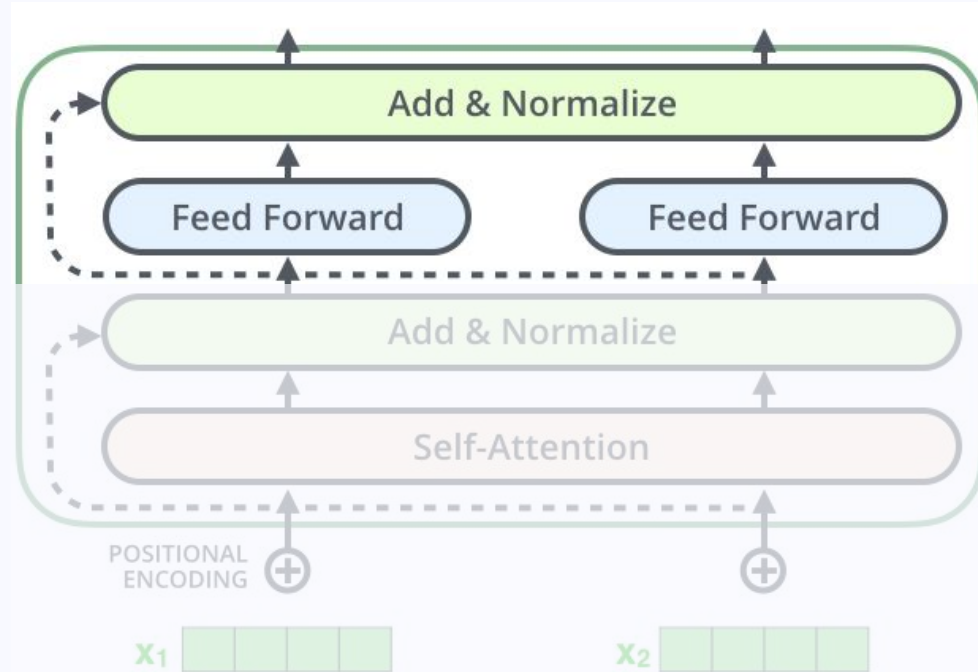
The Transformer's Encoder #1

1. Multi-headed Self-attention
2. Layer Normalization:
stabilize the hidden states



The Transformer's Encoder #1

1. **Multi-headed Self-attention**
2. **Layer Normalization:**
stabilize the hidden states
3. **Feedforward** neural network
4. More **layer normalization!**

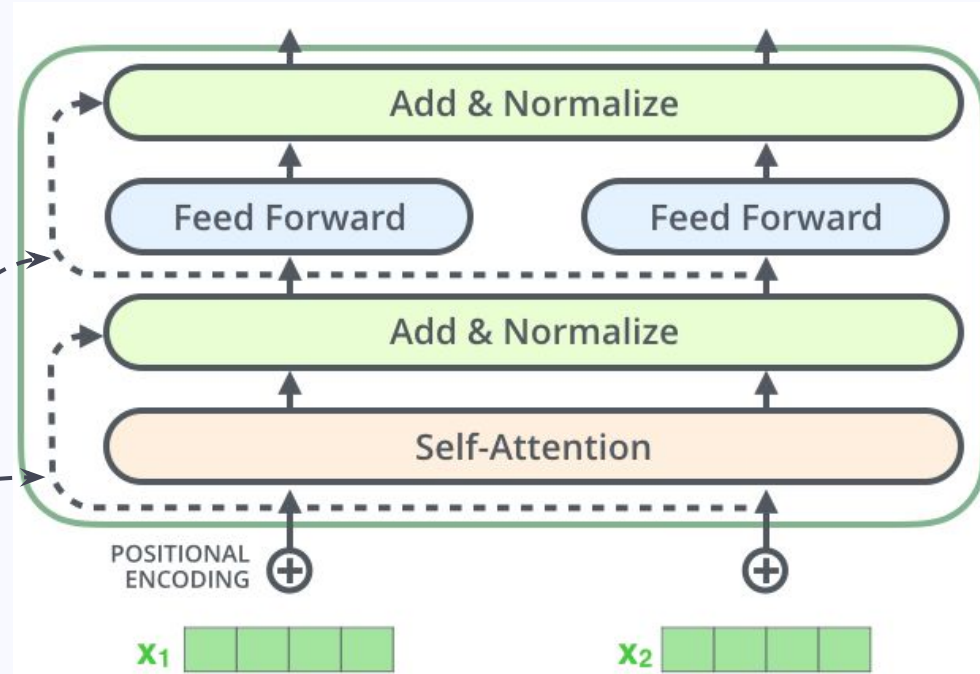


The Transformer's Encoder #1

1. **Multi-headed Self-attention**
2. **Layer Normalization:**
stabilize the hidden states
3. **Feedforward** neural network
4. More **layer normalization!**

Residual connection:

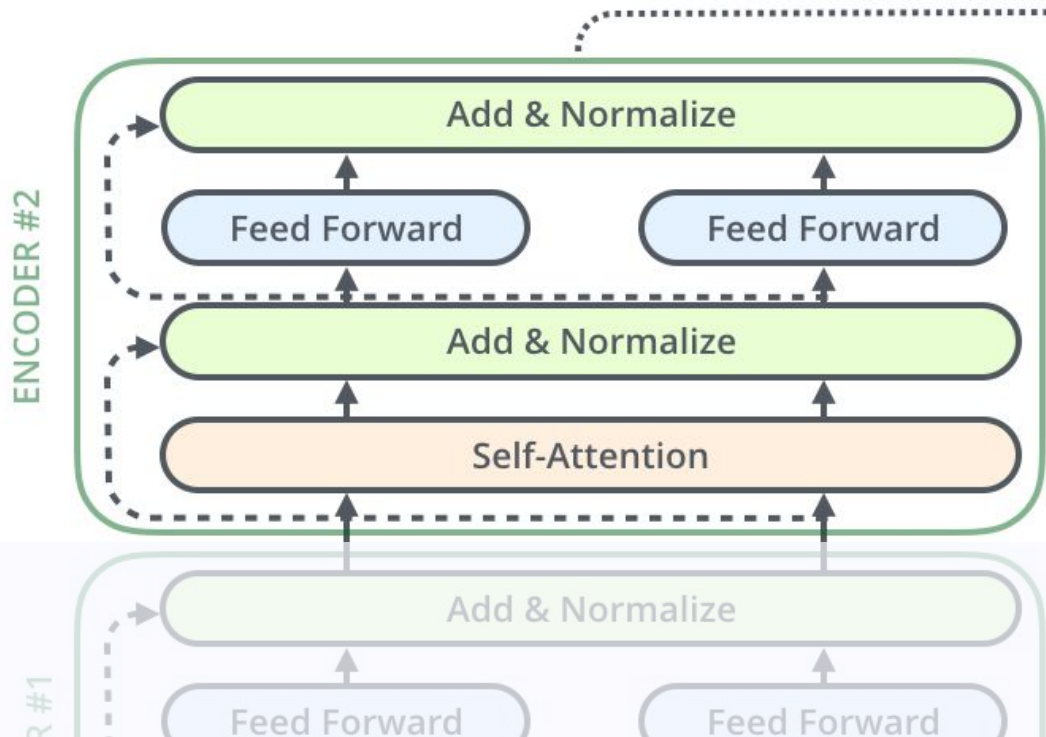
pass gradients through net →
no vanishing gradients!



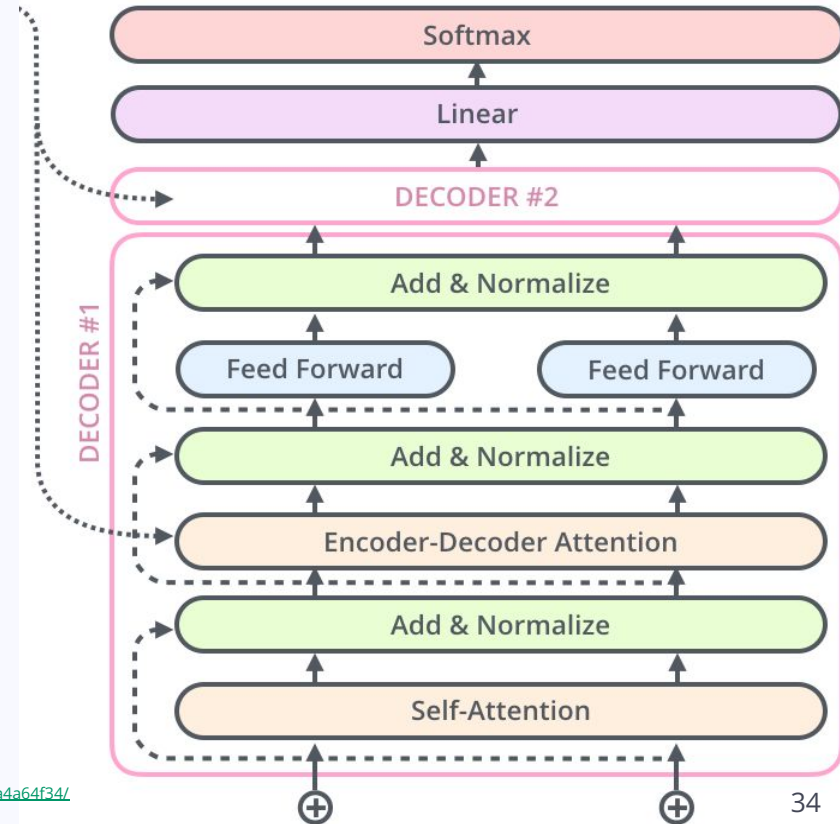
The Transformer's Encoder #2

Identical to Encoder #1!

(but receives Encoder #1's output, instead of the OG input sequence)

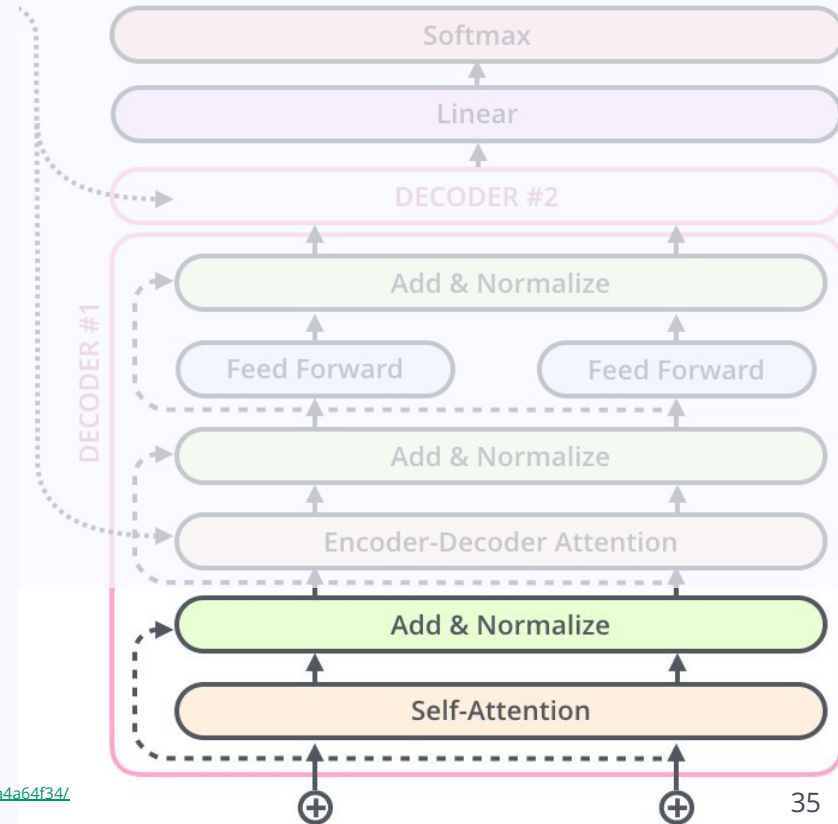


The Transformer's Decoder #1



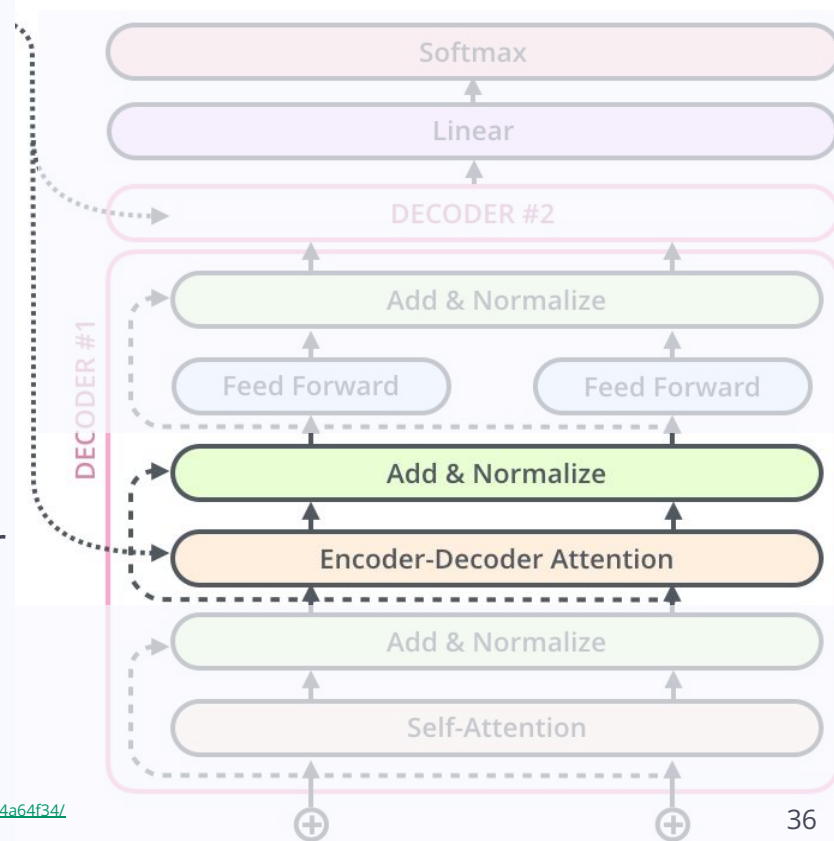
The Transformer's Decoder #1

1. Multi-head self-attention: **only** attend to earlier positions in the sequence
2. Layer normalization



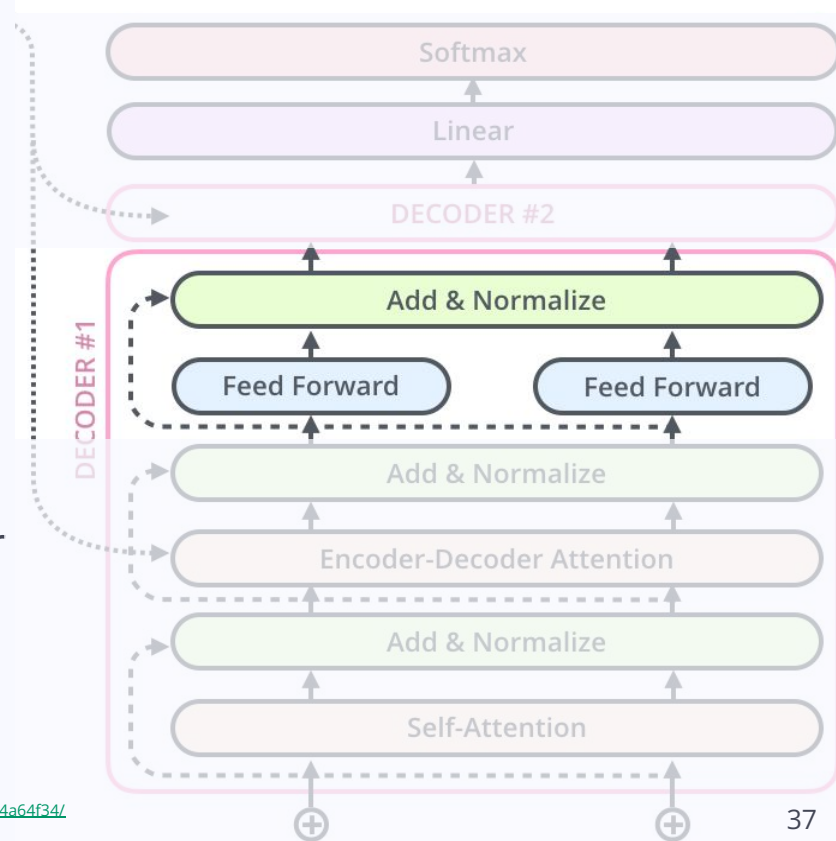
The Transformer's Decoder #1

1. Multi-head self-attention: **only** attend to earlier positions in the sequence
2. Layer normalization
3. **Encoder-decoder attention layer:** combine encoder output (.....)
+ output of previous self-attention layer



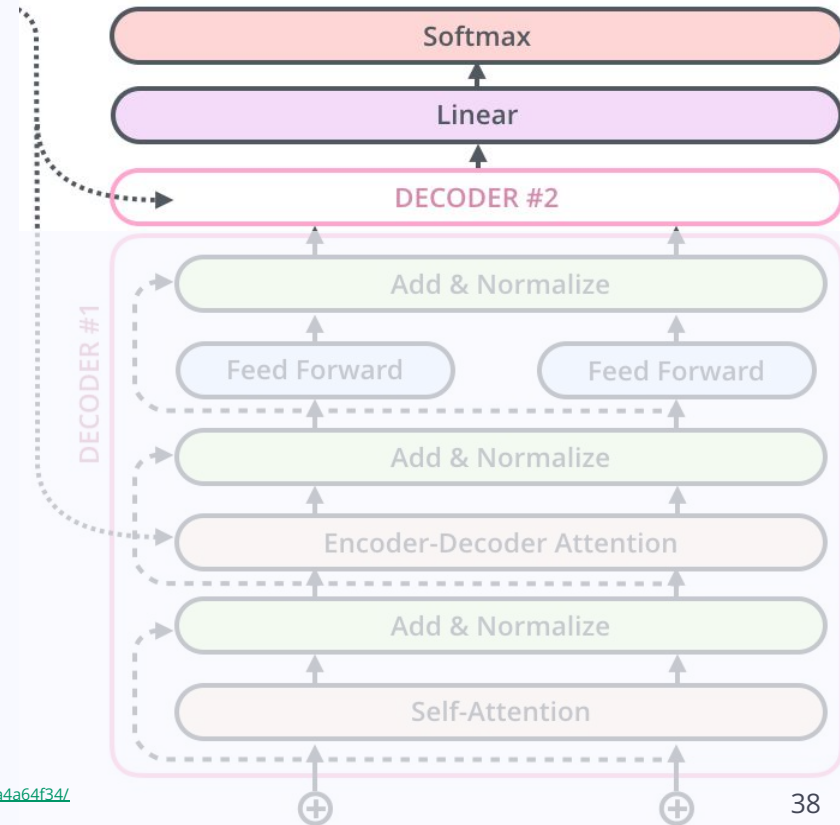
The Transformer's Decoder #1

1. Multi-head self-attention: **only** attend to earlier positions in the sequence
2. Layer normalization
3. **Encoder-decoder attention layer:** combine encoder output + output of previous self-attention layer
4. Feed-forward layers



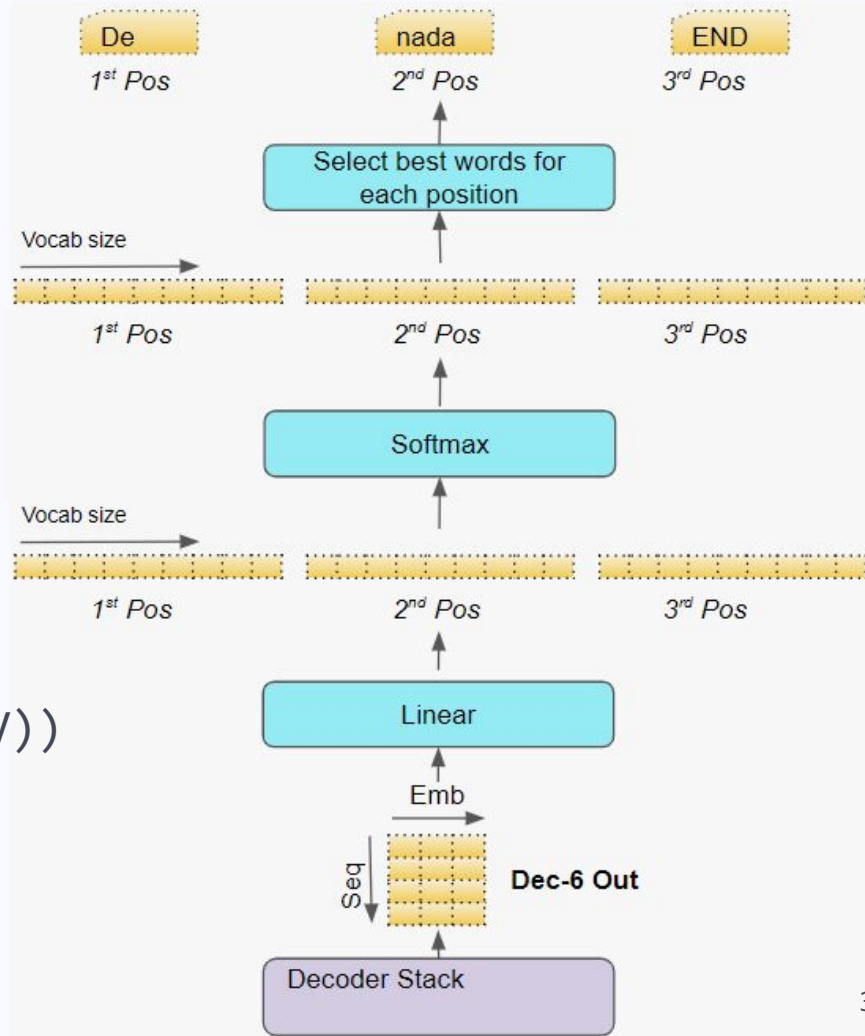
The Transformer's Decoder #2 etc.

- **Decoder #2:** Identical to Decoder #1
- **Linear layer:** assign probability to each token of the target vocabulary (logits)
- **Softmax layer:** turn probabilities 0-1 (log probabilities)
- **Output** = highest probability... maybe!



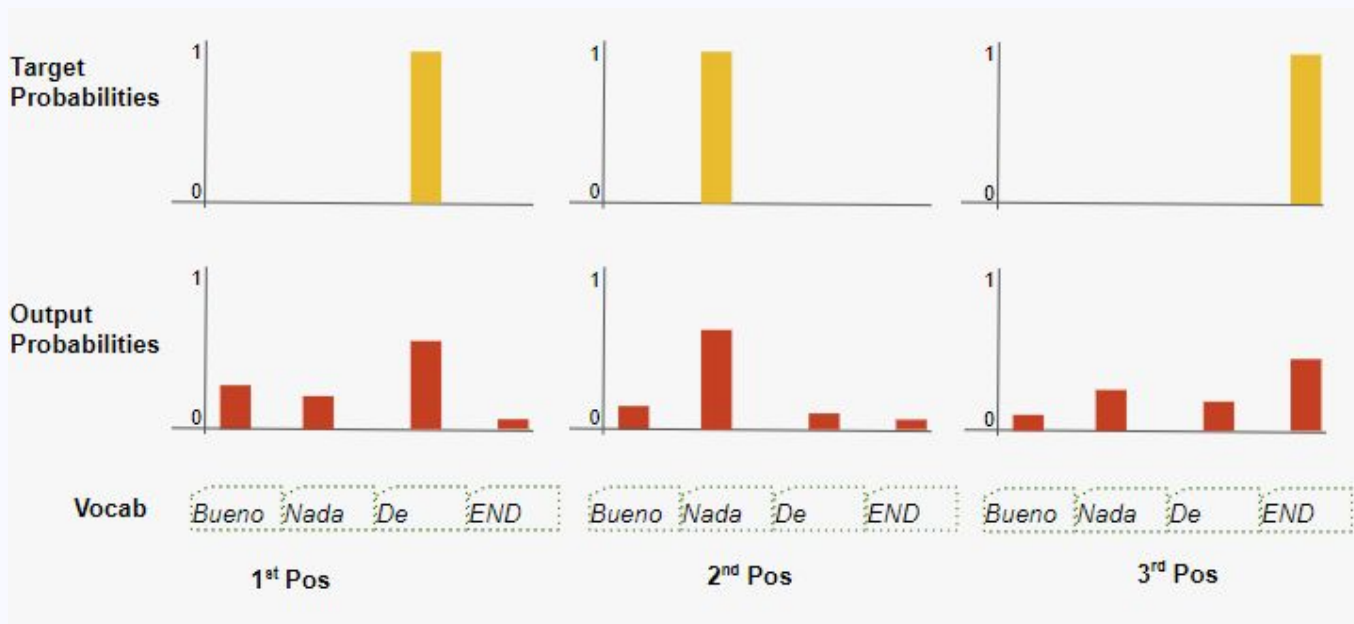
Generate Output

1. Decoder passes to Linear
2. Linear: for every word in V , score of word in each position
(logits: $(\text{len}(\text{output}), \text{len}(V))$)
3. Softmax: score $\rightarrow$ probability (0-1)
(log probs: $(\text{len}(\text{output}), \text{len}(V))$)



Training & Loss Function

We still use backpropagation to calculate loss!



3. (Large) Language Models

BERT

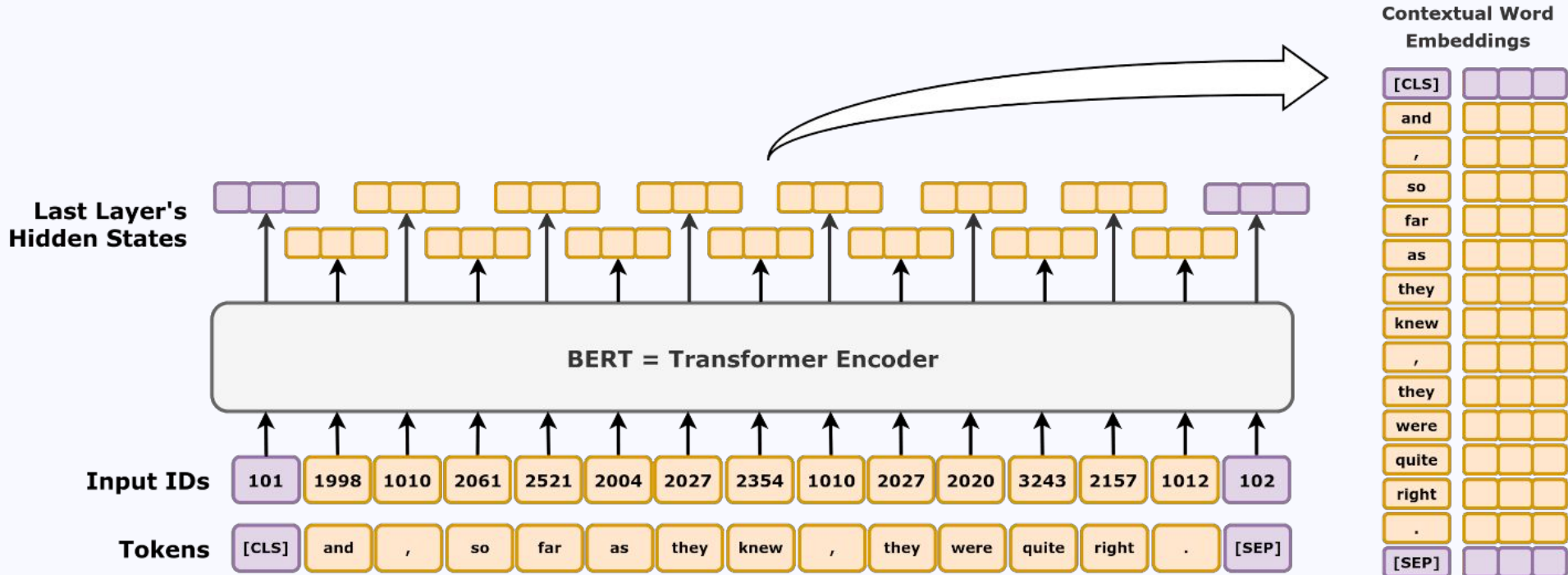
LLM spotlight: BERT



- **Bidirectional** : bidirectional self-attention!
- **Encoder** : its output = our context vectors!
- **Representations** : token embeddings relative to their context!
- from **Transformers** : uses a transformer NN!

Training goals: Masked Language Modeling & Next Sentence Prediction

Masked Language Modeling



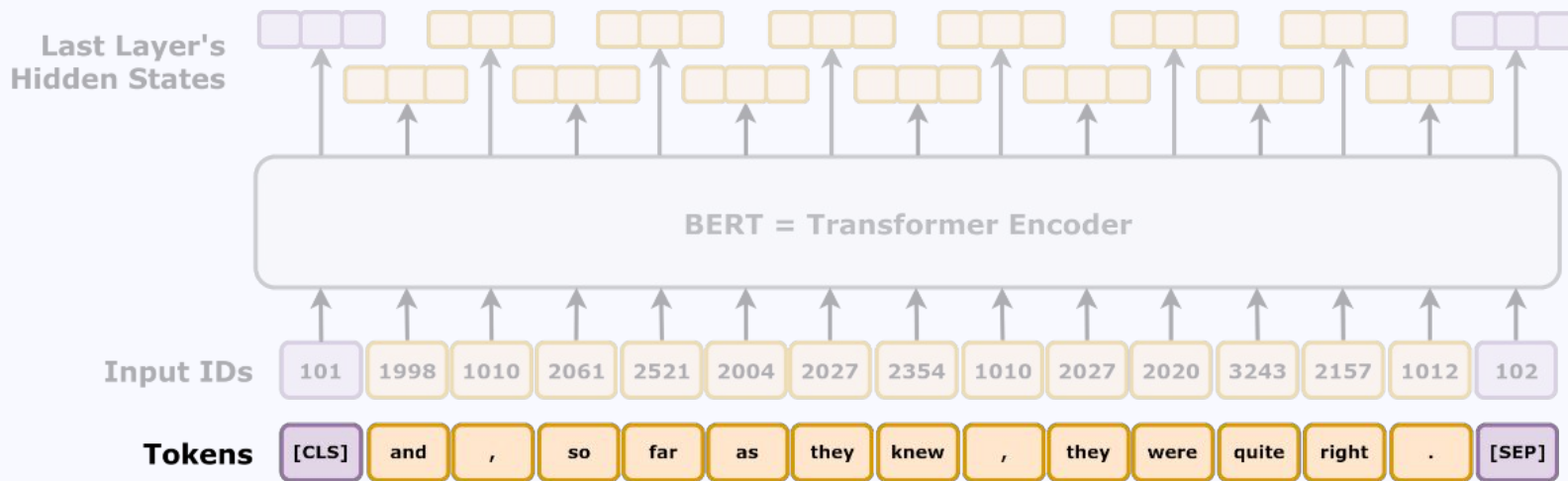
Masked Language Modeling

BERT's special tokens:

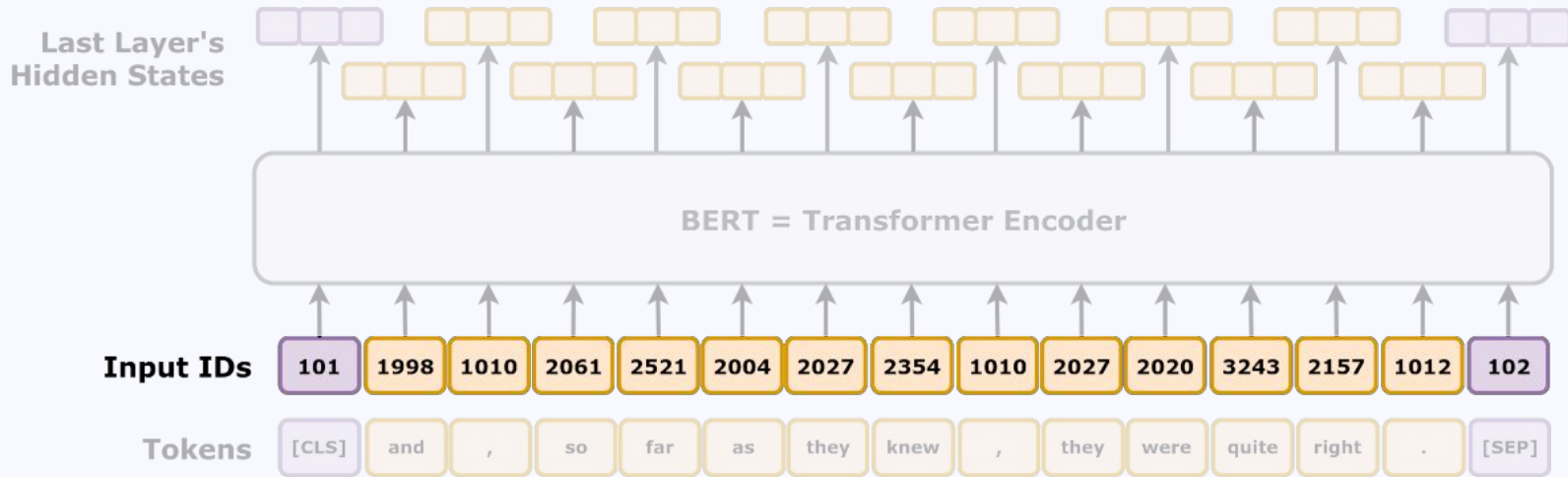
[CLS] = "Classify"

[SEP] = Separates phrases

[MASK] = Hides a token

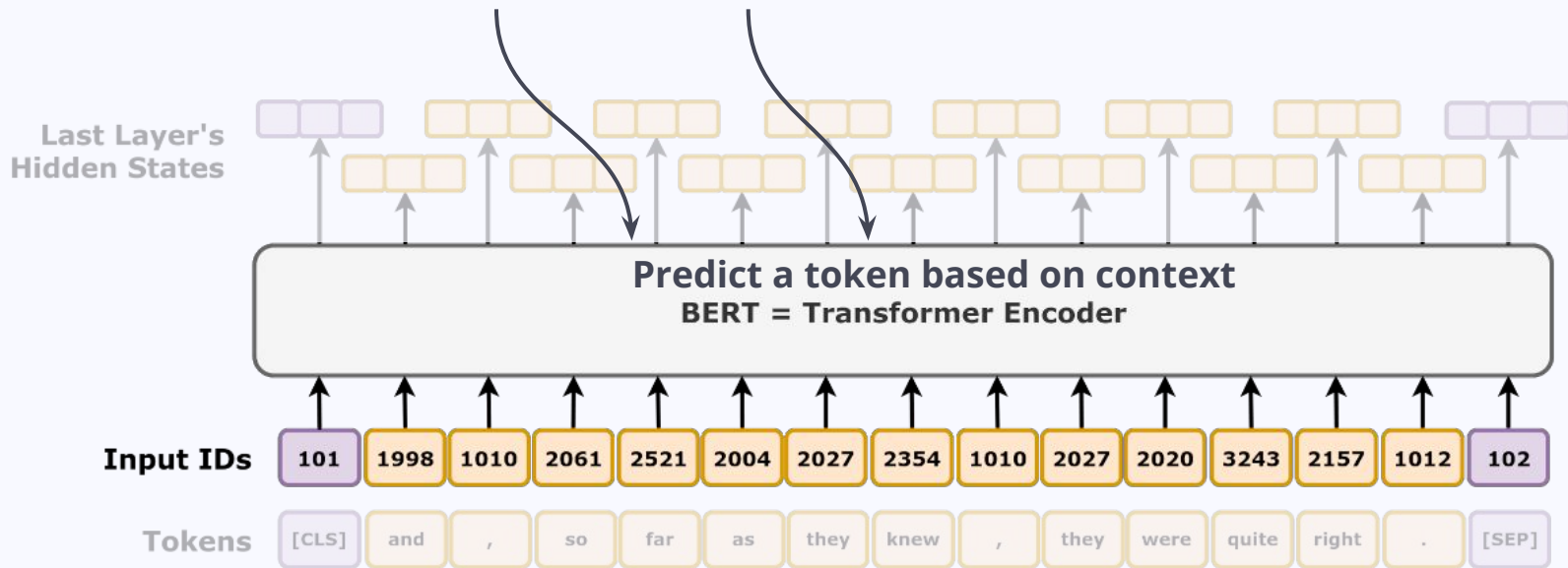


Masked Language Modeling

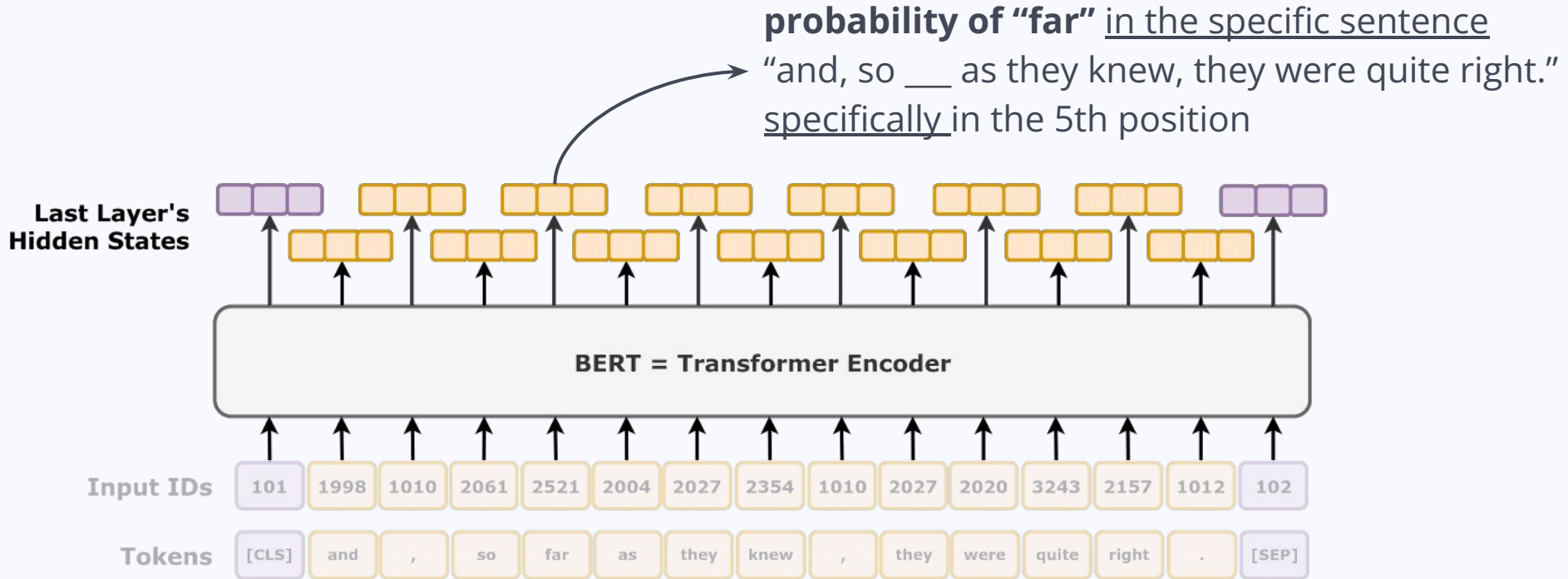


Masked Language Modeling

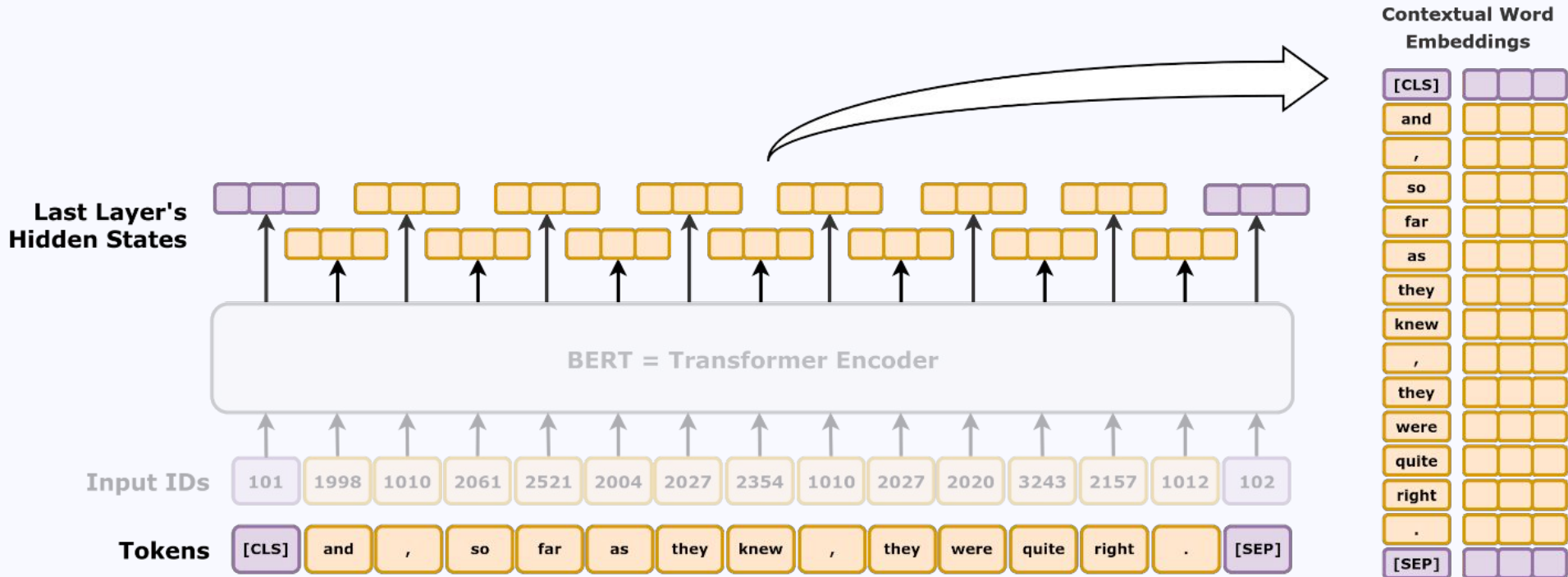
bert-base: **12** layers x **12** attention heads = 144 representations



Masked Language Modeling



Masked Language Modeling



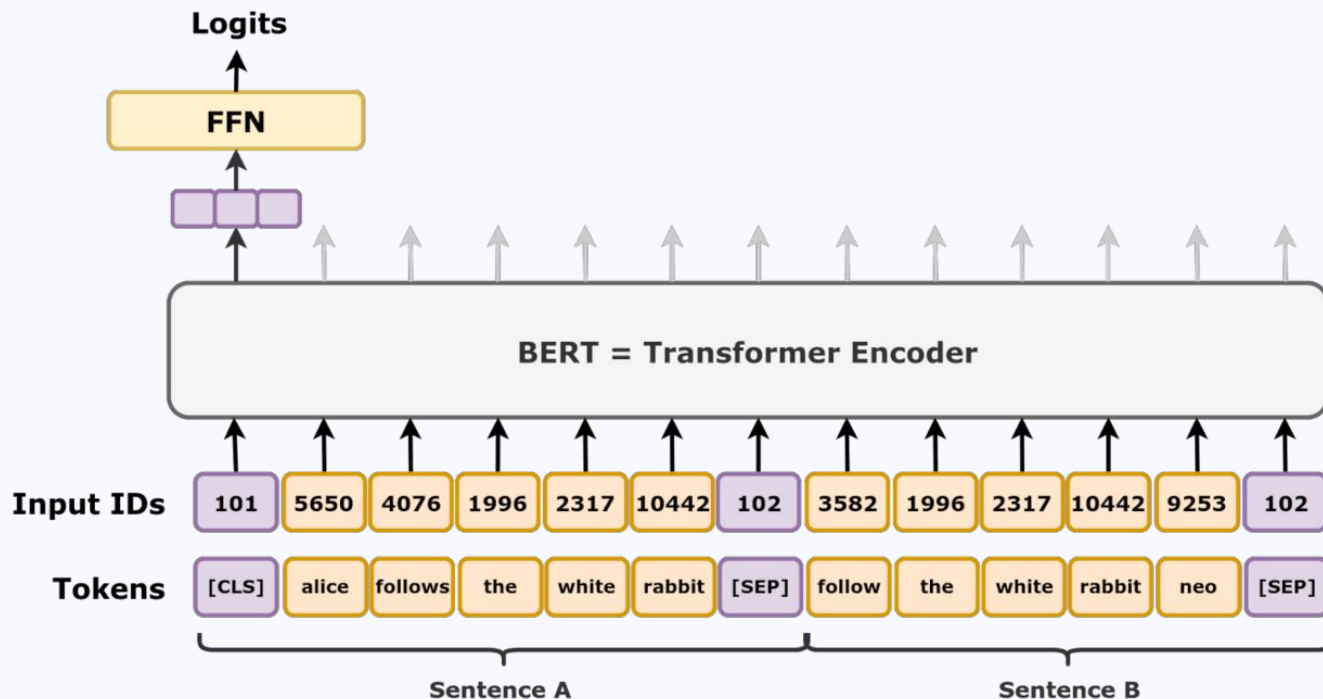
Next Sentence Prediction

BERT's special tokens:

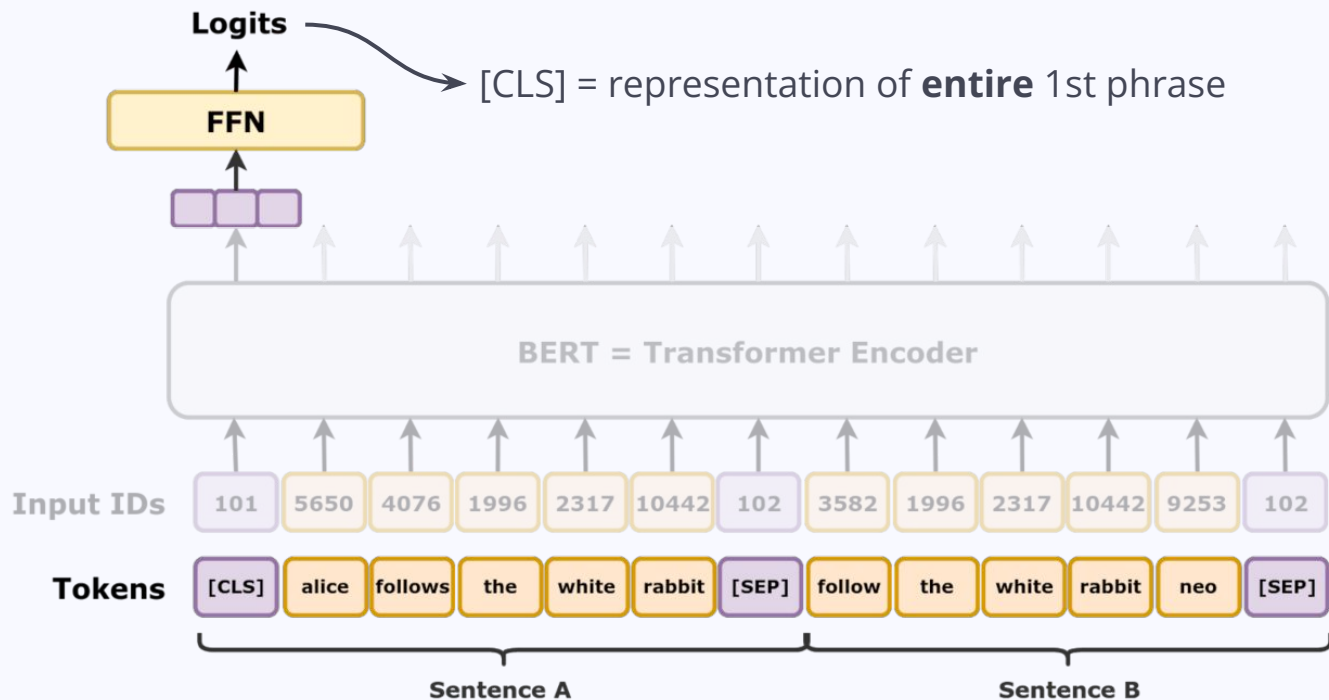
[CLS] = "Classify"

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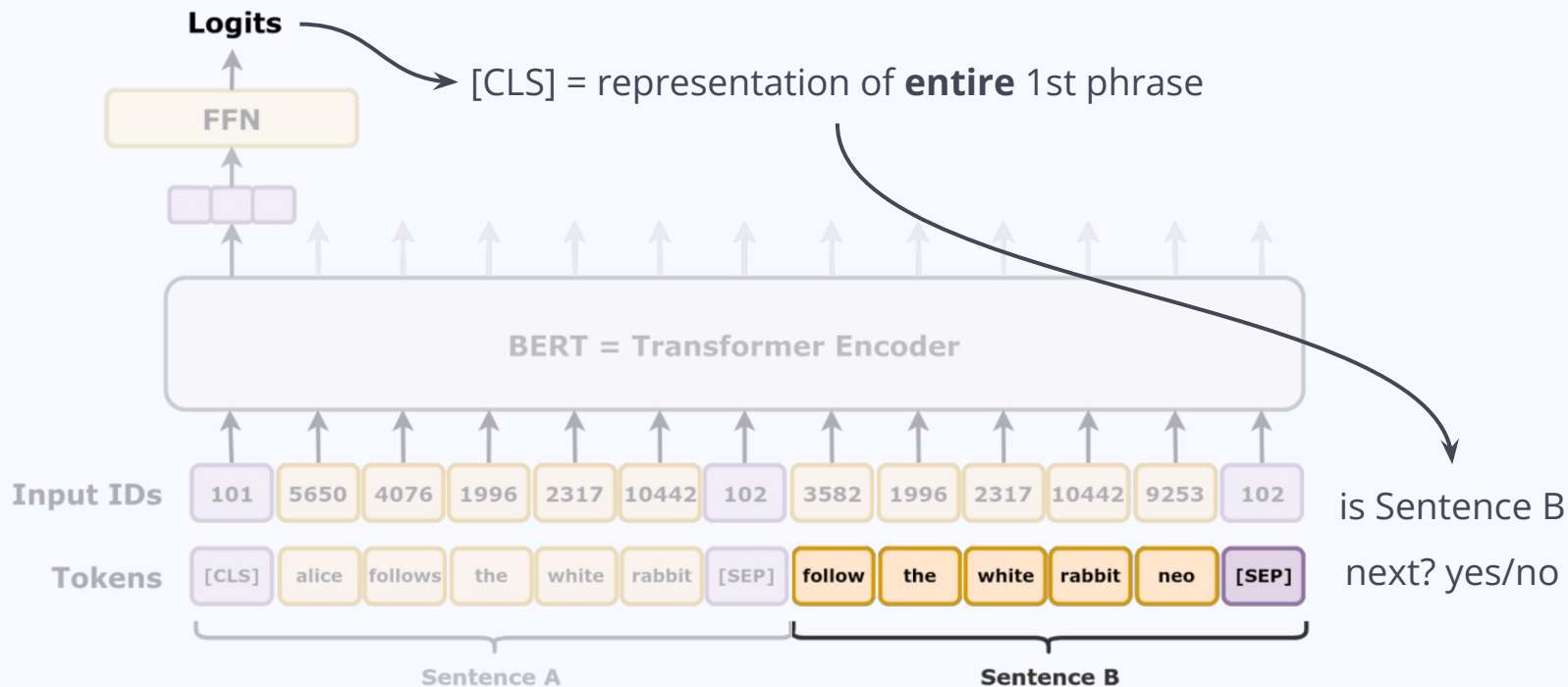
[MASK] = Hides a token



Next Sentence Prediction



Next Sentence Prediction



BERT representations

- [MASK] words to learn about them from context only!
- Cloze tasks: fill in the blanks!

I love eating ____ salad.

BERT representations

- [MASK] words to learn about them from context only!
- Cloze tasks: fill in the blanks!

I love eating ____ salad.

- spinach, Caesar, etc.
- fresh
- my mom's

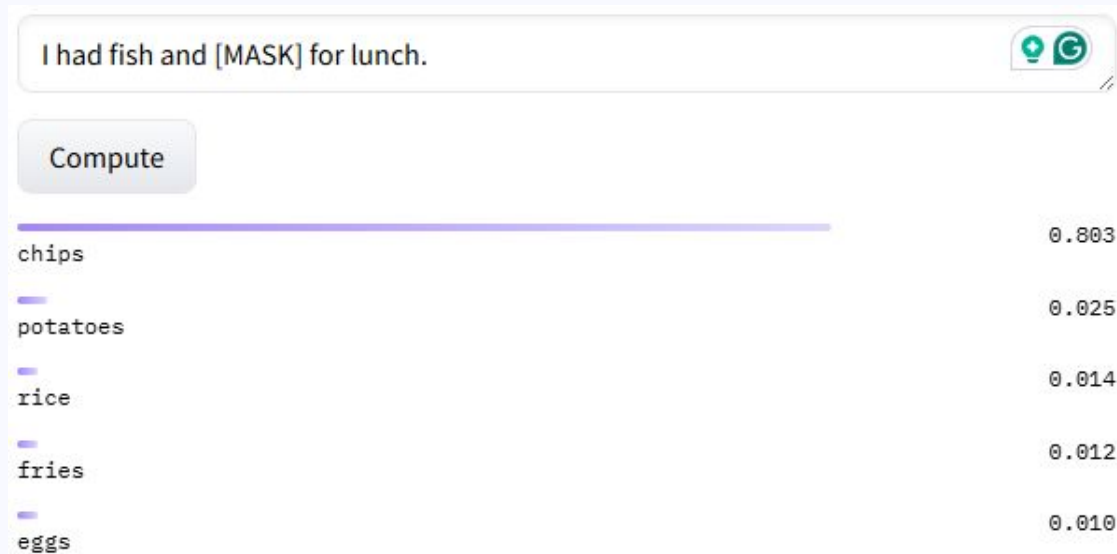


BERT representations

- [MASK] words to learn about them from context only!
- Cloze tasks: fill in the blanks!

I had fish and ____ for lunch.

“fish and chips” is a MWE!



The screenshot shows a web interface for a BERT model. At the top, there is a text input field containing "I had fish and [MASK] for lunch." with a green icon on the right. Below the input field is a "Compute" button. The results are displayed as a list of words with corresponding probability scores, separated by a horizontal line. The words are "chips", "potatoes", "rice", "fries", and "eggs". The scores are 0.803, 0.025, 0.014, 0.012, and 0.010 respectively. The word "chips" is highlighted with a purple bar.

Word	Score
chips	0.803
potatoes	0.025
rice	0.014
fries	0.012
eggs	0.010

What do BERT's embeddings know?

- Do they look like **traditional embeddings** (distribution, transformations)?
 - Yes... maybe in the higher layers
- Do they have **syntactic information**?
 - Bidirectionality really helped!
 - Parts of speech, syntactic chunks and roles, but not distant relations
 - No full syntactic trees, but syntactic transformations and dependencies
 - Bad with negation and with “bad” input

What do BERT's embeddings know?

- Do they have **semantic information**?
 - Some knowledge of semantic roles, entity types, relations, proto-roles
 - Can't generalize!
- Do they have **world knowledge**?
 - Fills the blanks successfully, but not enough!
 - Bad at inference, bias?!

BERT variations

- Bigger models: bert-large-cased
- Smaller models: distilbert-base-cased
- Improvements on training: roberta

and many, many more in different languages, sizes, tasks...

- BERT etc. models trained for specific NLP tasks: on [Hugging Face](#) 🙌!
[NER](#), [Question answering](#), [Summarization](#), [Sentiment classification](#), etc.
- BERT for [finance](#), [biology](#), etc. also on Hugging Face!

GPT

Transformer spotlight: GPT

- **Generative**: Auto-regression uses previous output sequence as input
- **Pretrained**: Just decoding!
- **Transformer**: Using Transformer blocks for decoding

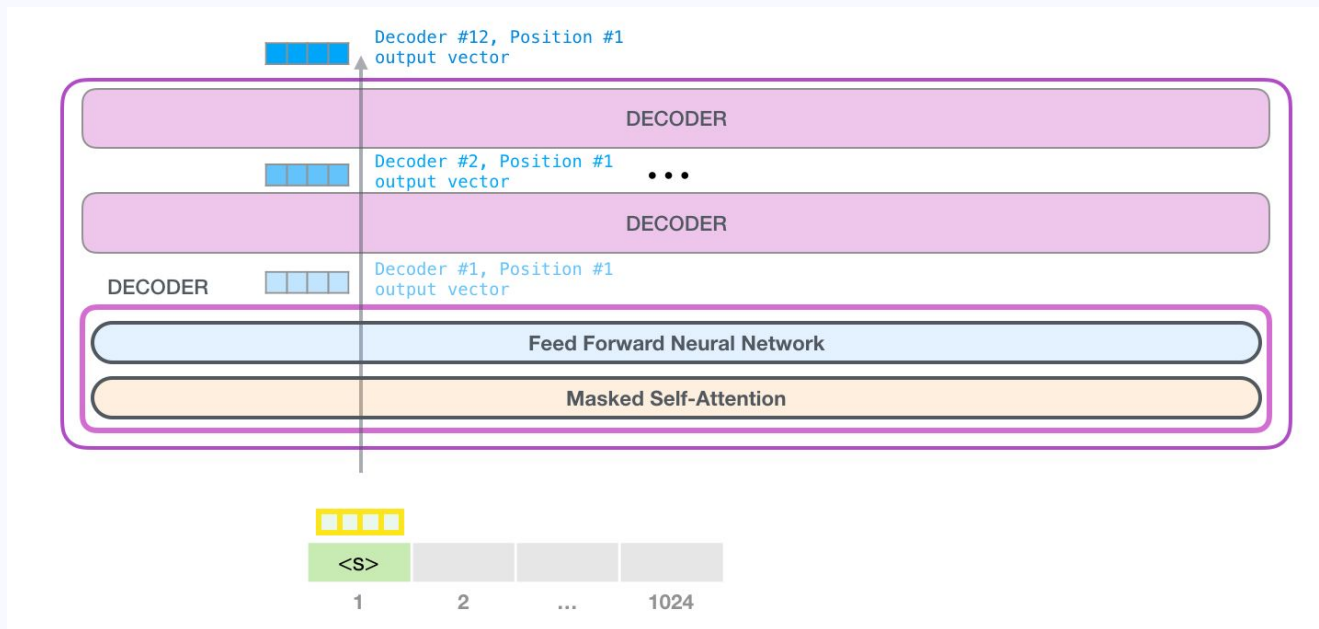
GPT Inputs

Token embeddings + Positional encodings = embedding of token X in position Y



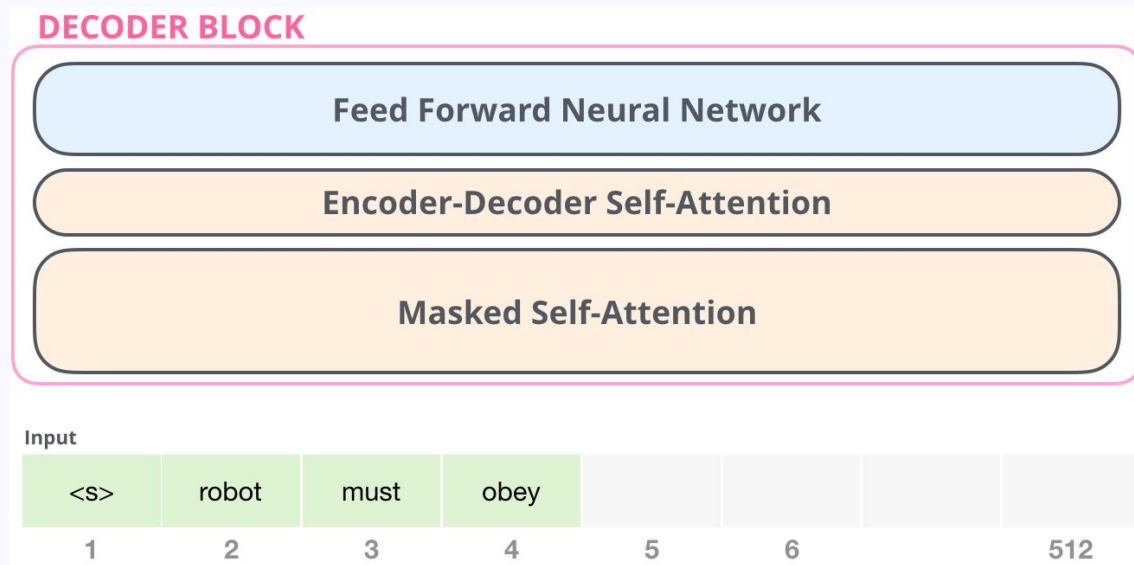
GPT: Through the Decoder Stacks

Each decoder processes the token, passes its output to the next decoder

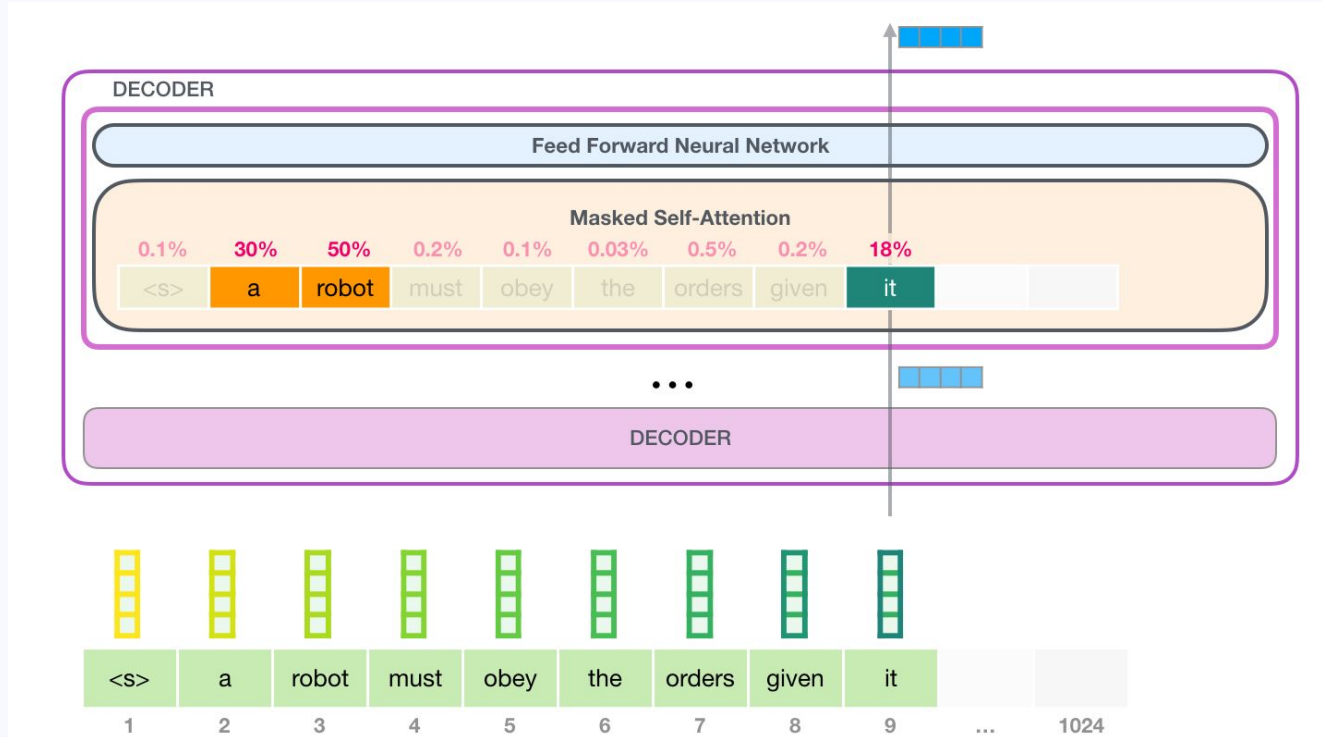


GPT Decoder block

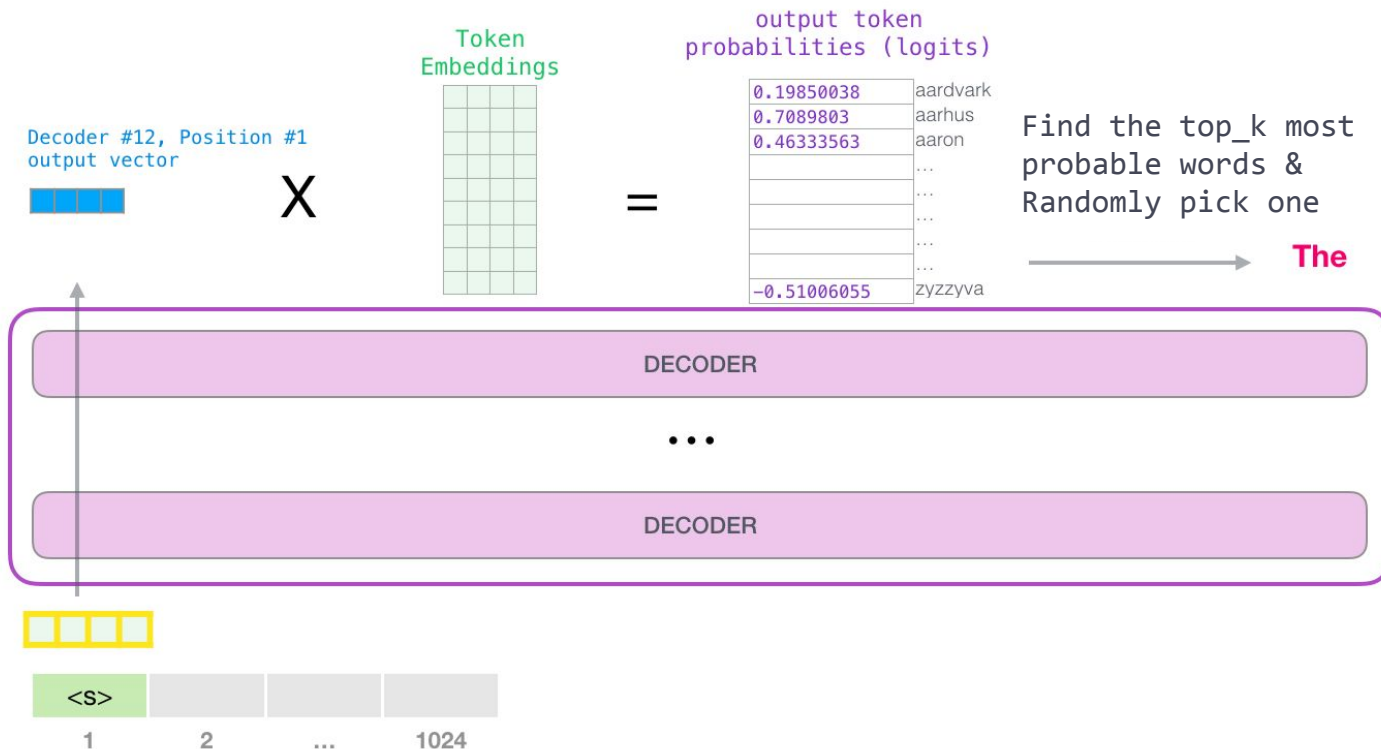
Masked self-attention in the decoder block = access only to the output's previously predicted tokens!



GPT Masked self-attention



GPT Output



Transformer Explainer: Visualize how GPT-2 processes input & generates next token

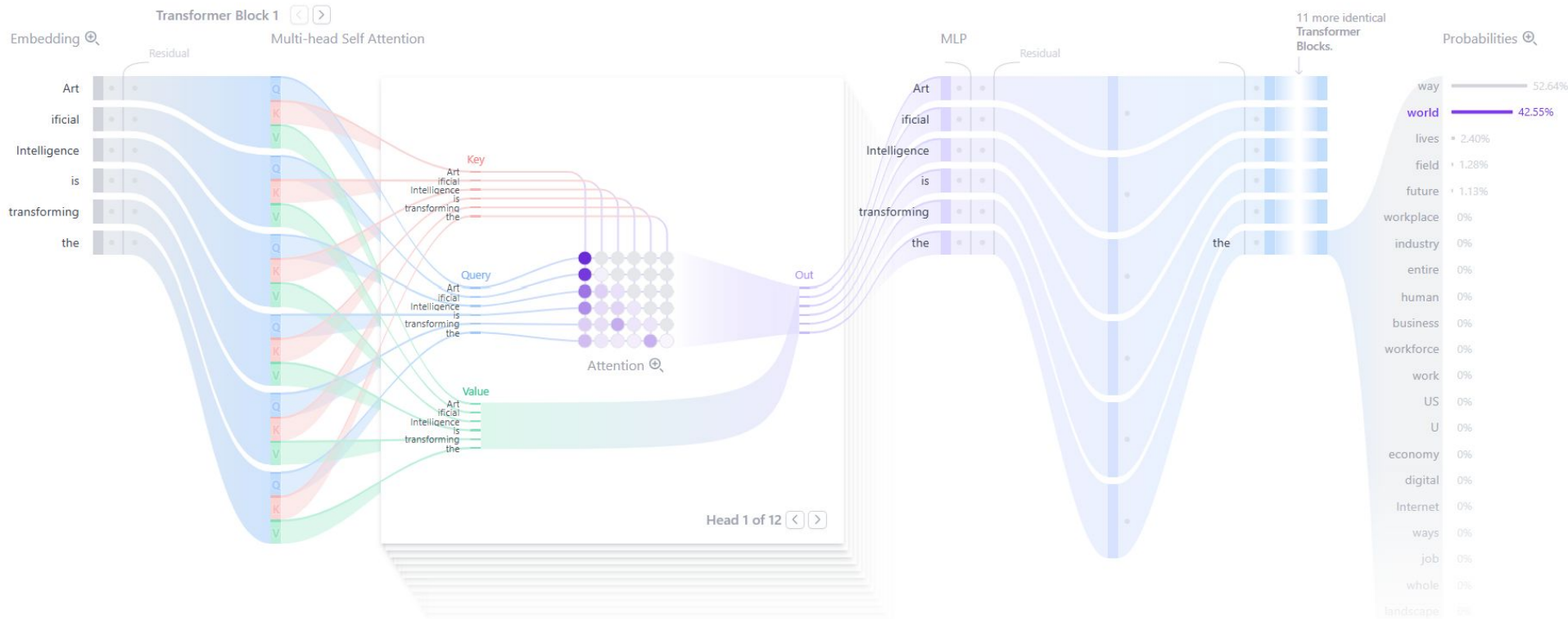
TRANSFORMER EXPLAINER

Examples ▾ Artificial Intelligence is transforming the world

Generate

Temperature 0.8

Sampling Top-k Top-p k=5





Chapter: Overview



GPT-2 (small)

nano-gpt

GPT-2 (XL)

GPT-3



nano-gpt
n_params = 85,584

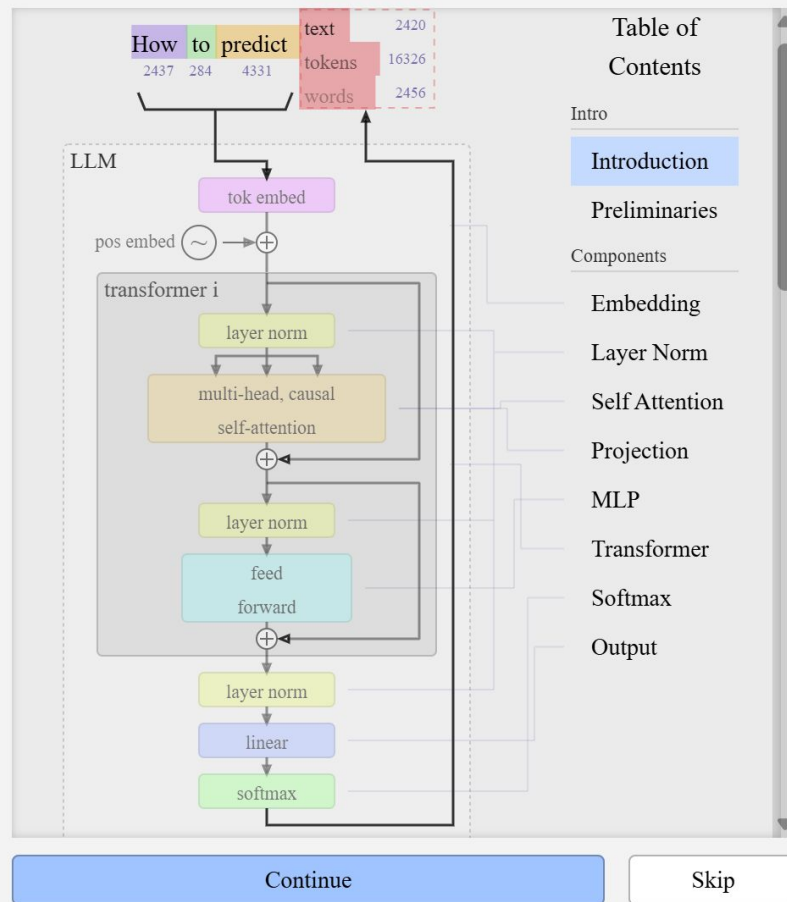


Table of Contents

Intro

Introduction

Preliminaries

Components

Embedding

Layer Norm

Self Attention

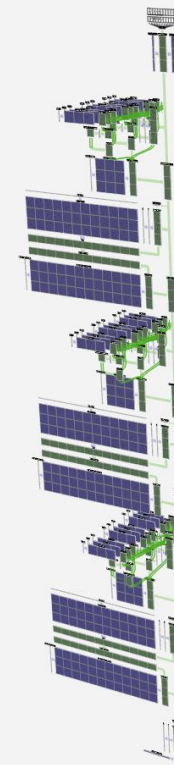
Projection

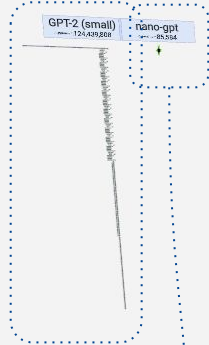
MLP

Transformer

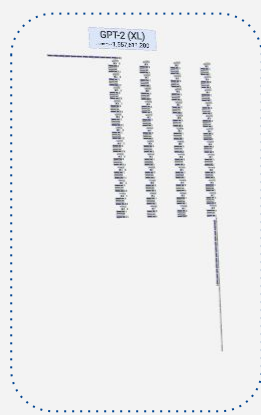
Softmax

Output



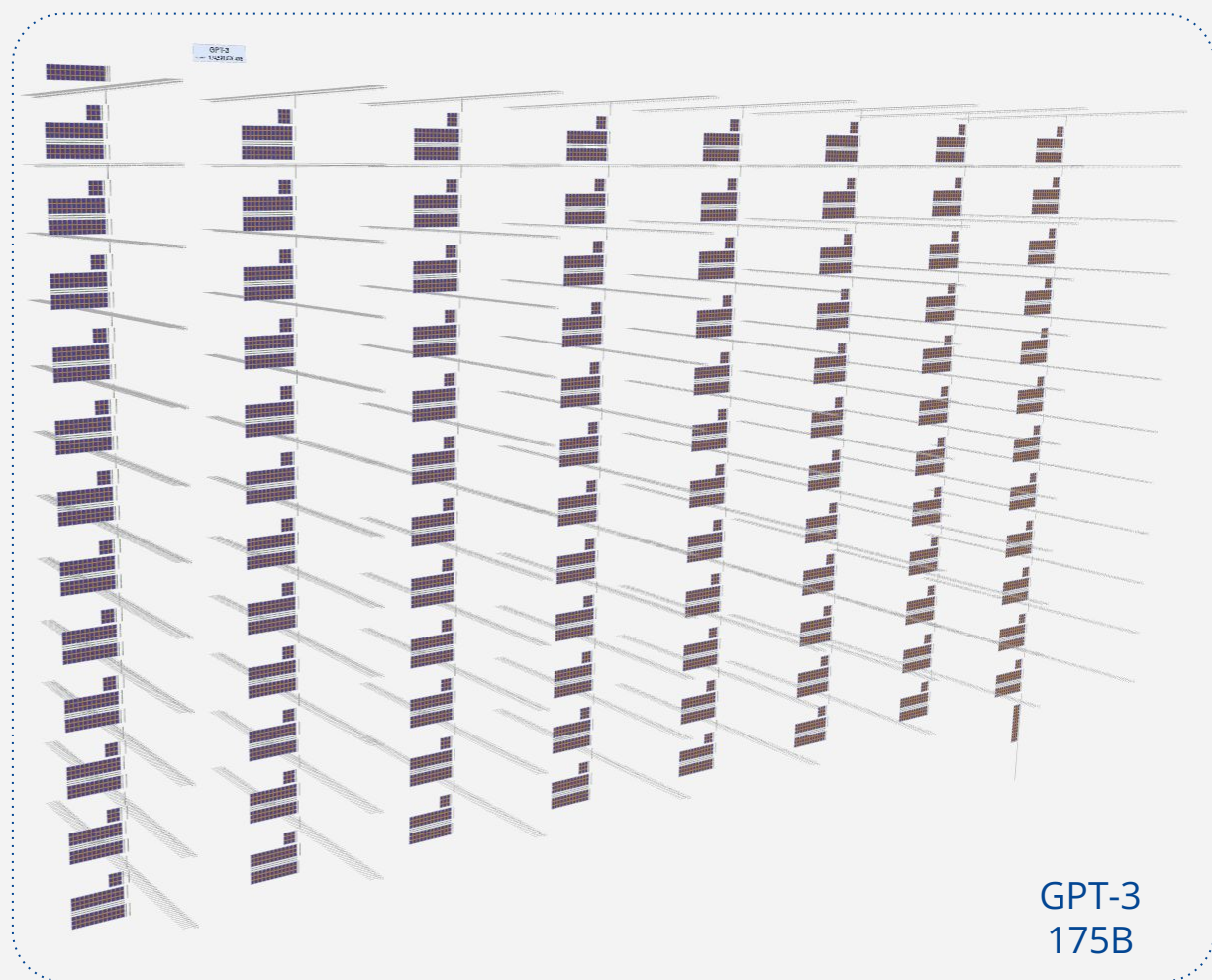


GPT-2
124M



GPT-2 XL
1.5B

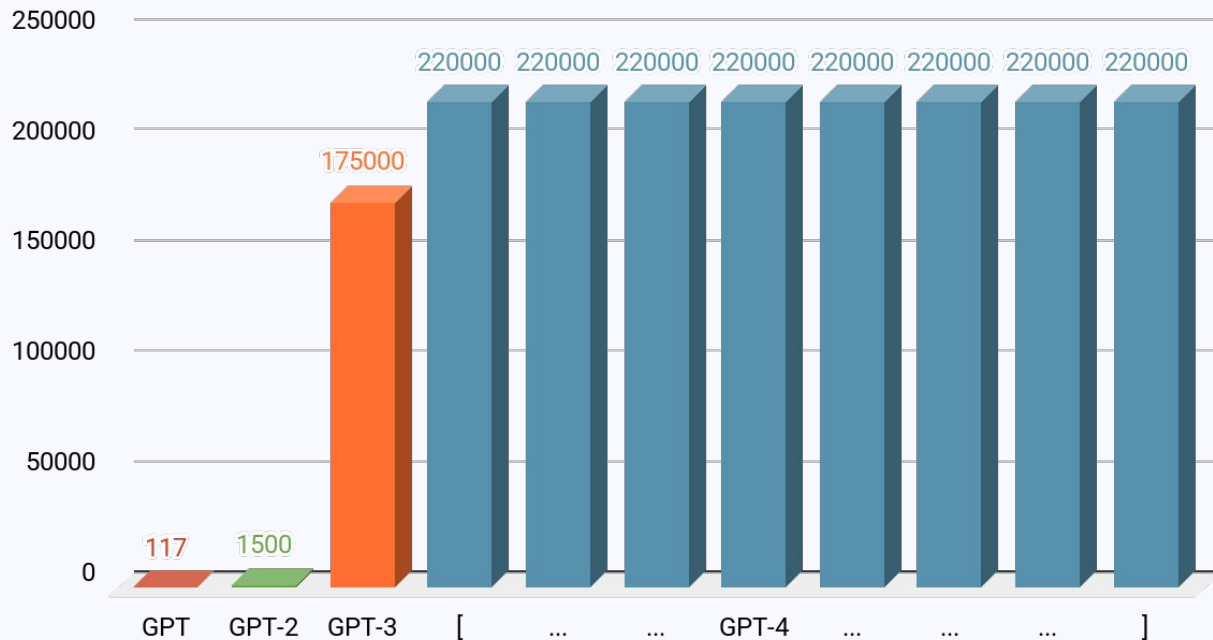
nano-GPT
85K



GPT-3
175B

GPT Model Sizes

Number of parameters (in millions)



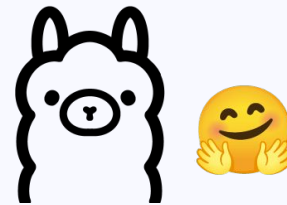
More LLMs

LLaMA models



- Also a **Transformer**-based architecture
- Bigger input embeddings (others: 512 dimensions, LLaMA: > 4.056 dims...)
- **Root Mean-Square normalization**: rescale vectors, not turn them ~ 0
- **Positional embeddings** in relation to the key-queries, not the token
- **Multi-query** self attention: same queries across attention heads
- LLaMA Instruct models: ideal for prompts that follow structured instructions

Mistral models

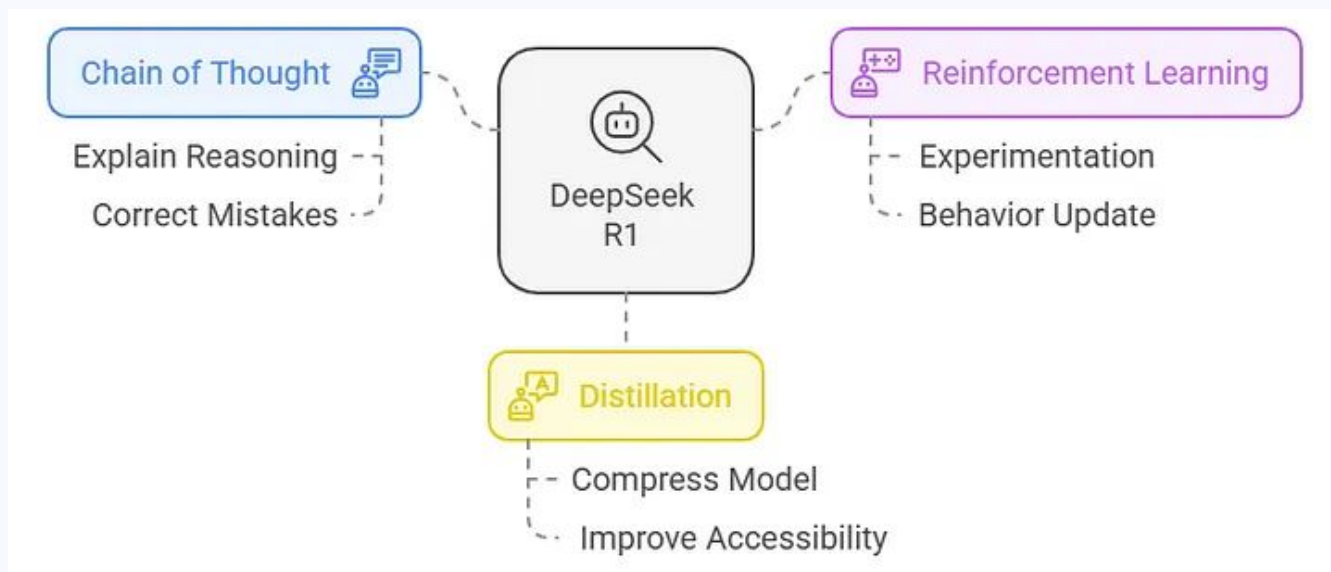


- Similar to LLaMa models
- **Optimizations** to improve memory use w.r.t. attention:
Sliding Window attention, Rolling Buffer Cache, Pre-fill and chunking

DeepSeek



Employing strategies to improve upon existing Transformer architectures



Improving LLMs

- **Distillation:** transferring knowledge from a large model to a smaller one with an algorithm or with generated data ($\neq$ pruning, $\neq$ compression)
- **Chain of Thought:** the model *has* to output the “thought” process of answer

Improving LLMs

- **Reinforcement Learning:** update the model with good answers to “forget” the bad ones
- **Supervised Finetuning:** finetune the model with high-quality outputs
- **Retrieval Augmented Generation:** ask the model to retrieve information from a specific source, not look up from its data

Hallucinations in LLMs

Unfaithful to the prompt:

- Doesn't follow instructions
- Doesn't care about context
- Doesn't follow a logical path

"Where is the Parthenon?"

Deepseek: "The Parthenon is located in Ancient France. It stands at an elevation of 66 meters (215 feet) and was constructed from the 10th to the 14th century under the rule of Charlemagne the Charpec. "

Contradictions and making up facts:

- Mentioning wrong/fake entities
- Making up fake relations
- Giving incomplete/outdated answers
- Giving unverifiable information
- Making up fake facts to fill an answer!

Practical Session

Preprocessing input

Preprocessing input

`from transformers import BertTokenizer` → BERT's specific tokenizer, with functions, BERT vocabulary and special tokens

Load the BERT tokenizer.

`print('Loading BERT tokenizer...')`

from pretrained =
load the model from HF

`tokenizer = BertTokenizer.from_pretrained('bert-base-uncased', do_lower_case=True)`

preprocessing option

bert-base-uncased = type of BERT

Print the original sentence.

`print(' Original: ', sentences[0])`

Print the sentence split into tokens.

`print('Tokenized: ', tokenizer.tokenize(sentences[0]))` → tokenize function

Print the sentence mapped to token ids.

`print('Token IDs: ', tokenizer.convert_tokens_to_ids(tokenizer.tokenize(sentences[0])))` → encode in BERT vocabulary

Loading BERT tokenizer...

Original: One more pseudo generalization and I'm giving up.

Tokenized: ['one', 'more', 'pseudo', 'general', '##ization', 'and', 'i', '', 'm', 'giving', 'up', '.']

Token IDs: [2028, 2062, 18404, 2236, 3989, 1998, 1045, 1005, 1049, 3228, 2039, 1012]

Preprocessing input

```
Loading BERT tokenizer...
```

```
Original: One more pseudo generalization and I'm giving up.
```

```
Tokenized: ['one', 'more', 'pseudo', 'general', '##ization', 'and', 'i', "'", 'm', 'giving', 'up', '.']
```

```
Token IDs: [2028, 2062, 18404, 2236, 3989, 1998, 1045, 1005, 1049, 3228, 2039, 1012]
```

long word: split in token + "function" subtoken!

...but what about the special tokens? Add them in the *encoding* stage!

```
input_ids = tokenizer.encode(sentences[1], add_special_tokens=True)
```

```
print('Token IDs: ', input_ids)
```

```
print('Tokenized: ', tokenizer.convert_ids_to_tokens(input_ids))
```

```
Token IDs: [101, 2028, 2062, 18404, 2236, 3989, 1998, 1045, 1005, 1049, 3228, 2039, 1012, 102]
```

```
Tokenized: ['[CLS]', 'one', 'more', 'pseudo', 'general', '##ization', 'and', 'i', "'", 'm', 'giving', 'up', '.', '[SEP]']
```


Preprocessing input

```
# `encode_plus` will:
#   (1) Tokenize the sentence.
#   (2) Prepend the `[CLS]` token to the start.
#   (3) Append the `[SEP]` token to the end.
#   (4) Map tokens to their IDs.
#   (5) Pad or truncate the sentence to `max_length`
#   (6) Create attention masks for [PAD] tokens.
encoded_dict = tokenizer.encode_plus(
    sent,                                # Sentence to encode.
    add_special_tokens = True,           # Add '[CLS]' and '[SEP]'
    max_length = 64,                    # Pad & truncate all sentences.
    pad_to_max_length = True,
    return_attention_mask = True,       # Construct attn. masks.
    return_tensors = 'pt',              # Return pytorch tensors.
)
```

Preprocessing input

Attention mask: tokens to “show to/hide from” the model (for MLM)

Input		I	like	[MASK]	salad	.	
Tokens	[CLS]	i	like	[MASK]	salad	.	[SEP]
Encoded tokens	101	1045	2066	103	16521	1012	102
Attention mask	1	1	1	0	1	1	1

Finetuning

Finetuning

The process of adapting a pre-trained model for specific tasks or use cases

- Parameters of existing model + New data with specific topic/interest
- Either on the entire network or on some layers or in a new final layer
- **Goal:** specialize the model on a new task with few data, but also use the power of the model's learned representations
- Save up time and computational power!

Finetuning

with PyTorch

```
# For each epoch...
for epoch_i in range(0, epochs):

    # =====
    #           Training
    # =====

    # Perform one full pass over the training set.

    print("")
    print('==== Epoch {:} / {:} ====='.format(epoch_i + 1, epochs))
    print('Training...')

    # Measure how long the training epoch takes.
    t0 = time.time()

    # Reset the total loss for this epoch.
    total_train_loss = 0

    # Put the model into training mode. Don't be misled--the call to
    # `train` just changes the *mode*, it doesn't *perform* the training.
    # `dropout` and `batchnorm` layers behave differently during training
    # vs. test (source: https://stackoverflow.com/questions/51433378/what-d)
    model.train()
```

with HuggingFace

```
training_args = TrainingArguments(
    output_dir="my_awesome_model",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    weight_decay=0.01,
    evaluation_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    push_to_hub=True,
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_imdb["train"],
    eval_dataset=tokenized_imdb["test"],
    tokenizer=tokenizer,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

trainer.train()
```

Language Generation

How to generate the best text?

- **Greedy search:** always return the top probability
- **Beam search:** “remember” other top probabilities, return the top probability
- **Sampling:** allow our model to pick words randomly... or randomly from the top ones!
- **Temperature:** control the smoothness of the probability distribution (0-1)
 = pick token with highest prob. /  = make token probs more uniform
- **top_k:** how many tokens from the highest ranking ones are to be considered
- **top_p:** how many possible words to consider

Thank you for your (self) attention!

Google Colab:

https://colab.research.google.com/drive/1SDC7Fqll-BSpUne8CeCtK_iDlyqVW8S6?usp=sharing

Further reading

- Reading:
 - [The Illustrated Transformer](#) - Jay Alammar
 - [The Illustrated BERT, ELMo, and co.](#) - Jay Alammar
 - [The Illustrated GPT-2](#) - Jay Alammar
 - [Illustrated Guide to Transformers- Step by Step Explanation](#) - Michael Phi
- Video Courses:
 - Jay Alammar's course: [How transformer LLMs work](#)
 - StatsQuest
- Notebooks:
 - [Tensor2Tensor intro](#): Colab Notebook with visualizations
 - [More notebooks folder](#): look at embeddings, run BERT, prompt LLMs, do RAG!